

Supporting Information

Selective production of light α -olefins and long-chain α -olefins from CO₂/H₂ and CO/H₂ over iron-based catalysts: effects of Na₂S and H₂O

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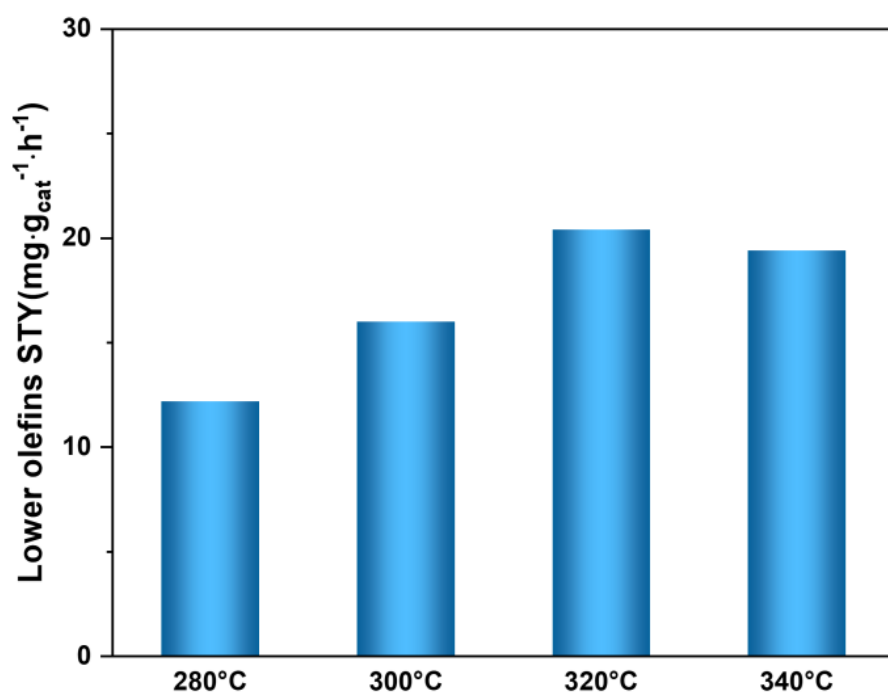


Figure S1. Calculated lower olefins STY values for CO₂-FTS reaction over 1.5Na-Fe1Zn1 catalyst under different reaction temperature.

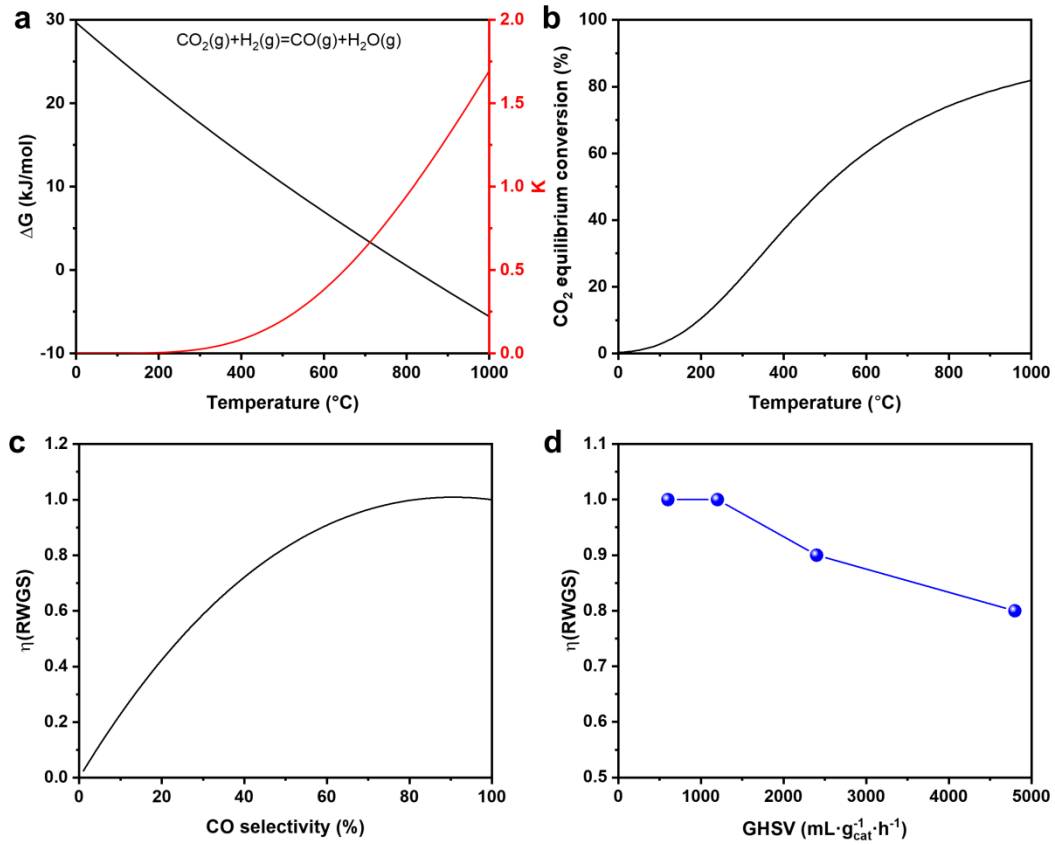


Figure S2. Thermodynamic analysis on RWGS reaction. (a) Reaction Gibbs free energy and equilibrium constant, (b) CO_2 equilibrium conversion, (c) relationship between CO selectivity and equilibrium degree of RWGS reaction at 320 °C by fixing CO_2 conversion at 25.8%, and (d) calculated equilibrium degree of RWGS reaction over 1.5Na-Fe1Zn1 catalyst under different GHSV.

In order to elucidate the promoting effect of CO hydrogenation into hydrocarbons on CO_2 conversion, we first supposed an RWGS reaction, along with CO_2 equilibrium conversion of 25.8% and CO selectivity of 100%. Then, CO hydrogenation ($\text{CO} + 2\text{H}_2 \rightarrow \text{CH}_2^* + \text{H}_2\text{O}$) occurred leading to the change in CO selectivity, H_2 and H_2O partial pressure. We supposed that CO_2 conversion was fixed at 25.8% and then established a correlation between CO selectivity and $\eta(\text{RWGS})$ as shown in Figure S2c. It can be seen that $\eta(\text{RWGS})$ significantly decreased when CO selectivity decreased into about 0%-20%. Therefore, CO_2 tended to undergo the forward reaction, leading to the increase in CO_2 conversion. It suggested that the rapid consumption of CO in CO_2 -FTS reaction could promote CO_2 conversion.

In order to obtain the flow rate of i under reaction conditions, CO_2 conversion, H_2 conversion and CO selectivity (X_{CO_2} , X_{H_2} and S_{CO}) were calculated as follows:

$$X_{\text{CO}_2} = \frac{(A_{\text{CO}_2})_{\text{in}}/(A_{\text{N}_2})_{\text{in}} - (A_{\text{CO}_2})_{\text{out}}/(A_{\text{N}_2})_{\text{out}}}{(A_{\text{CO}_2})_{\text{in}}/(A_{\text{N}_2})_{\text{in}}}$$

$$X_{\text{H}_2} = \frac{(A_{\text{H}_2})_{\text{in}}/(A_{\text{N}_2})_{\text{in}} - (A_{\text{H}_2})_{\text{out}}/(A_{\text{N}_2})_{\text{out}}}{(A_{\text{H}_2})_{\text{in}}/(A_{\text{N}_2})_{\text{in}}}$$

$$S_{\text{CO}} = \frac{f_{\text{CO}}(A_{\text{CO}})_{\text{out}}/(A_{\text{N}_2})_{\text{out}}}{f_{\text{CO}_2}[(A_{\text{CO}_2})_{\text{in}}/(A_{\text{N}_2})_{\text{in}} - (A_{\text{CO}_2})_{\text{out}}/(A_{\text{N}_2})_{\text{out}}]}$$

where, $(A_i)_{in}$ and $(A_i)_{out}$ represented for the peak area of i in GC spectrum into and out of the reactor, respectively, and f_{CO} and f_{CO_2} represented for the calibrated factor of CO and CO_2 , respectively. Subsequently, the flow rate of i out of the reactor was calculated using the following equations:

$$\begin{aligned}(F_{CO_2})_{out} &= (F_{CO_2})_{in} \times (1 - X_{CO_2}) \\ (F_{H_2})_{out} &= (F_{H_2})_{in} \times (1 - X_{H_2}) \\ (F_{CO})_{out} &= (F_{CO_2})_{in} \times X_{CO_2} \times S_{CO} \\ (F_{H_2O})_{out} &= (F_{CO_2})_{in} \times X_{CO_2} \times 2 - (F_{CO})_{out}\end{aligned}$$

where, $(F_i)_{in}$ represented for the flow rate of i into the reactor.

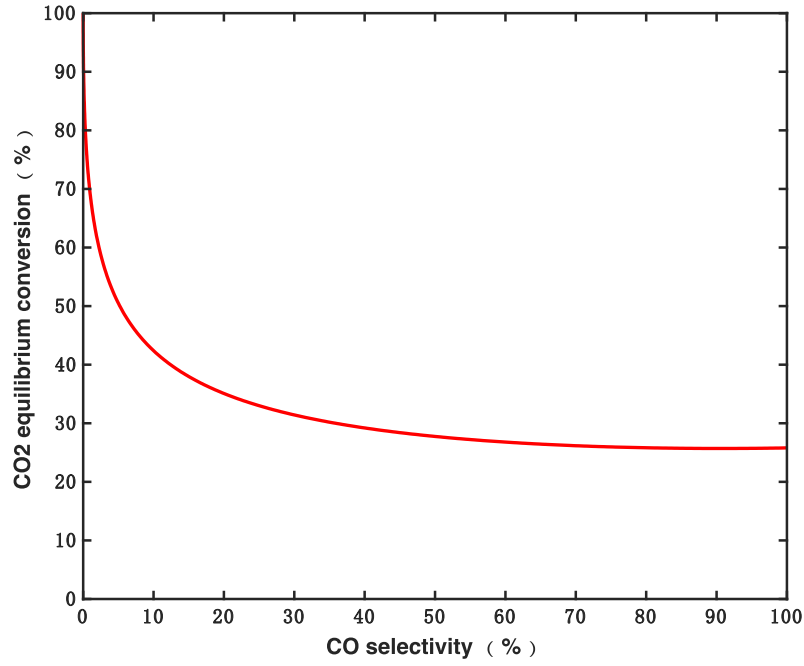


Figure S3. The relationship between CO selectivity and CO₂ equilibrium conversion.

In order to elucidate the relationship between CO selectivity and CO₂ equilibrium conversion, we first supposed an RWGS reaction, along with CO₂ conversion of $y\%$ and CO selectivity of 100%. Then, CO hydrogenation ($\text{CO} + 2\text{H}_2 \rightarrow \text{CH}_2^* + \text{H}_2\text{O}$) occurred leading to the change in CO selectivity from 100% to $x\%$. We supposed at this time, the RWGS reaction was at equilibrium. Therefore, the flow rate of CO₂, CO, H₂ and H₂O, and their correlation could be calculated as follows:

$$\begin{aligned}
 (F_{\text{CO}_2})_{\text{out}} &= (F_{\text{CO}_2})_{\text{in}} \times (1 - y\%) \\
 (F_{\text{CO}})_{\text{out}} &= (F_{\text{CO}_2})_{\text{in}} \times y\% \times x\% \\
 (F_{\text{H}_2\text{O}})_{\text{out}} &= (F_{\text{CO}_2})_{\text{in}} \times y\% + (F_{\text{CO}_2})_{\text{in}} \times y\% \times (1 - x\%) \\
 (F_{\text{H}_2})_{\text{out}} &= (3 - y\%) \times (F_{\text{CO}_2})_{\text{in}} - 2 \times y\% \times (1 - x\%) \times (F_{\text{CO}_2})_{\text{in}} \\
 \frac{(F_{\text{H}_2\text{O}})_{\text{out}} \times (F_{\text{CO}})_{\text{out}}}{(F_{\text{CO}_2})_{\text{out}} \times (F_{\text{H}_2})_{\text{out}}} &= K
 \end{aligned}$$

Based on the above equations, the relationship between CO selectivity and CO₂ equilibrium conversion was established and shown in Figure S3.

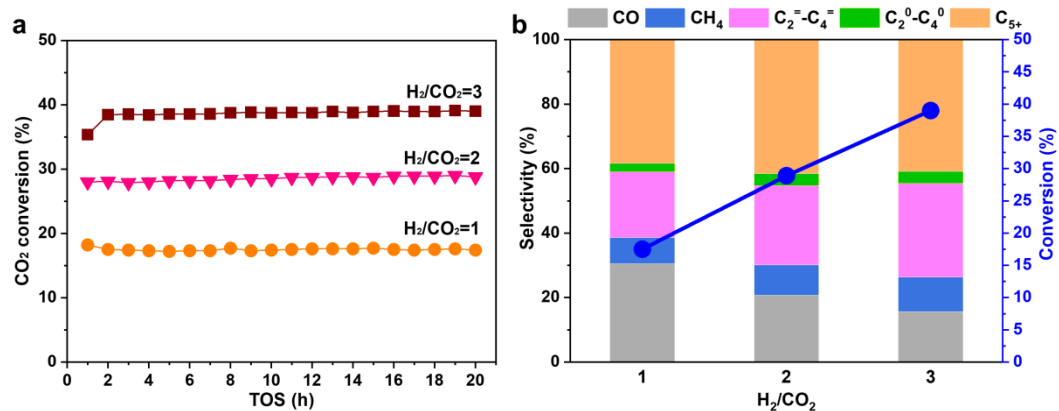


Figure S4. Catalytic performance of the 1.5Na-Fe1Zn1 catalyst for CO₂-FTS reaction. Effects of the H₂/CO₂ on the (a) catalytic activity and (b) product distribution. Reaction conditions: 2 g catalyst, 320 °C, 1 MPa, and GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

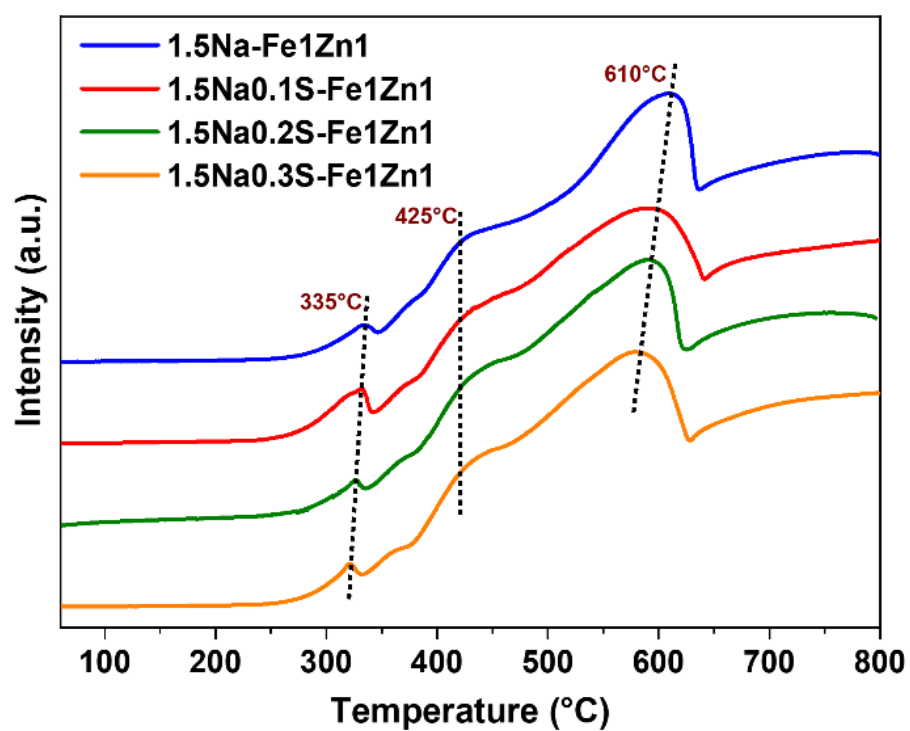


Figure S5. H₂-TPR profiles of the fresh NaS-Fe1Zn1 catalysts with different Na₂S content.

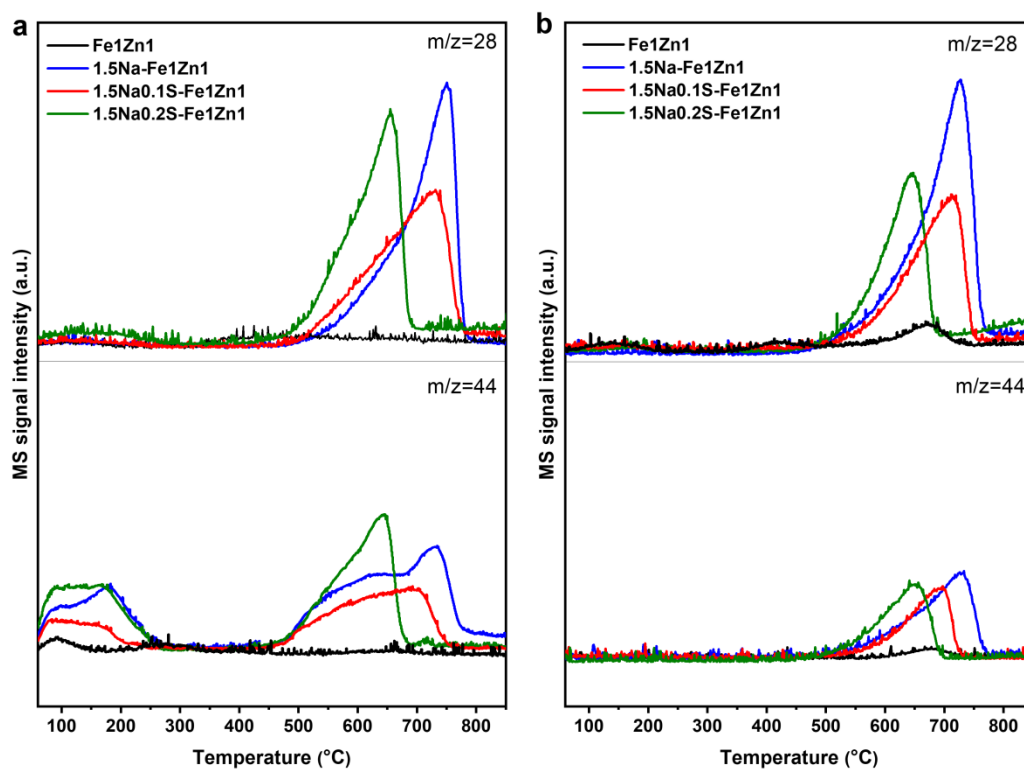


Figure S6. (a) CO₂-TPD and (b) CO-TPD profiles of the fresh NaS-Fe₁Zn₁ catalysts with different Na₂S content.

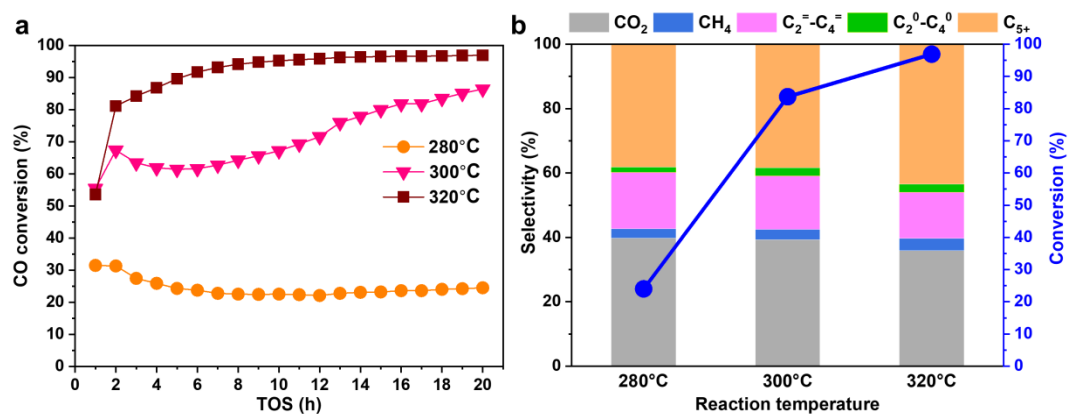


Figure S7. Catalytic performance of the 1.5Na_{0.1}S-Fe₁Zn₁ catalyst for CO-FTS reaction. Effects of the reaction temperature on the (a) catalytic activity and (b) product distribution. Reaction conditions: 2 g catalyst, H₂/CO = 1, 1 MPa, and GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

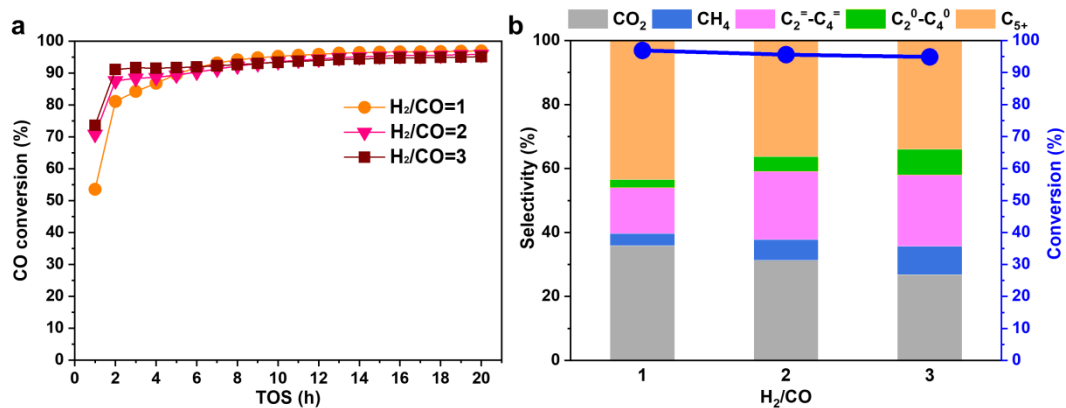


Figure S8. Catalytic performance of the 1.5Na0.1S-Fe1Zn1 catalyst for CO-FTS reaction. Effects of the H₂/CO ratio on the (a) catalytic activity and (b) product distribution. Reaction conditions: 2 g catalyst, 1 MPa, 320 °C, and GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

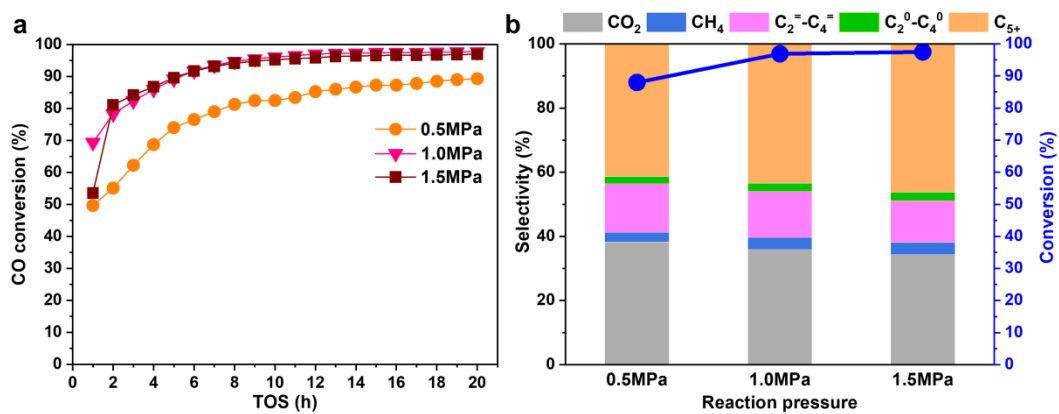


Figure S9. Catalytic performance of the 1.5Na_{0.1}S-Fe₁Zn₁ catalyst for CO-FTS reaction. Effects of the reaction pressure on the (a) catalytic activity and (b) product distribution. Reaction conditions: 2 g catalyst, H₂/CO = 1, 320 °C, and GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

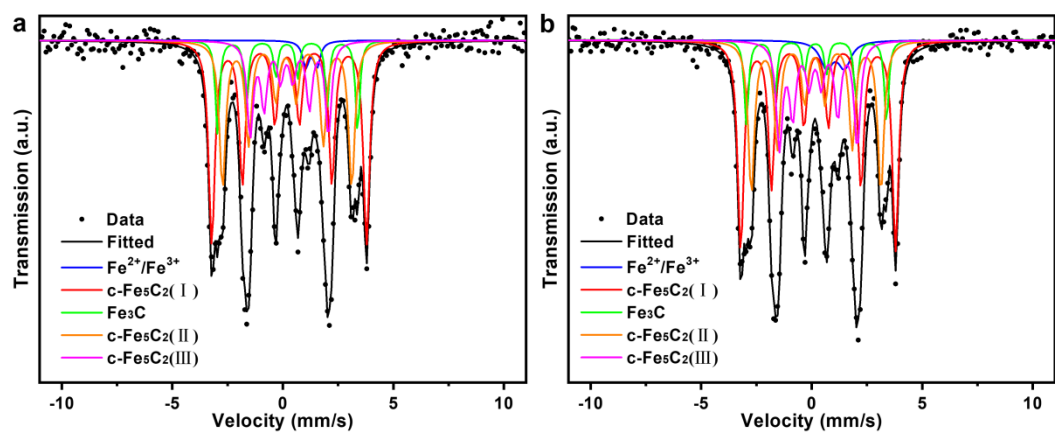


Figure S10. Mössbauer spectroscopy of the spent (a) 1.5Na-Fe1Zn1 and (b) 1.5Na0.1S-Fe1Zn1 catalyst after CO-FTS reaction.

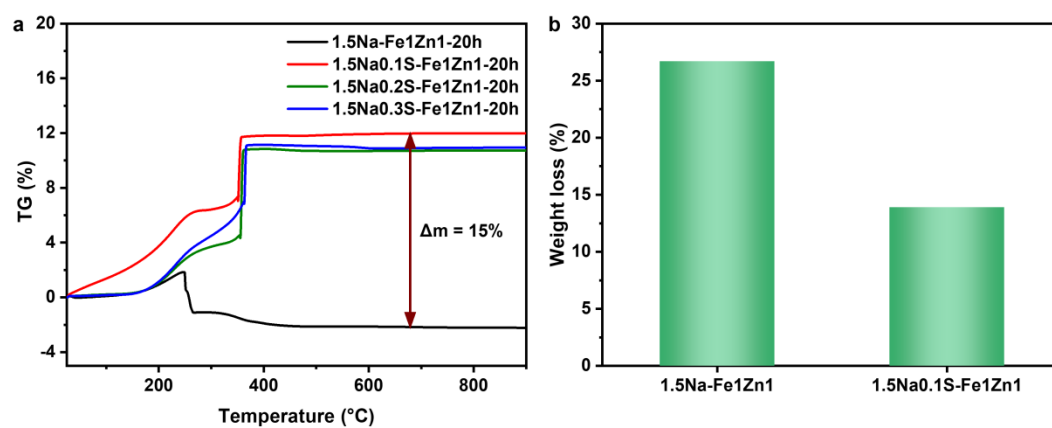


Figure S11. (a) TG profiles and (b) calculated weight loss of the spent NaS-Fe1Zn1 catalysts with different Na₂S content after CO-FTS reaction. The calculation of weight loss was based on the TG profiles and Mössbauer spectroscopy.

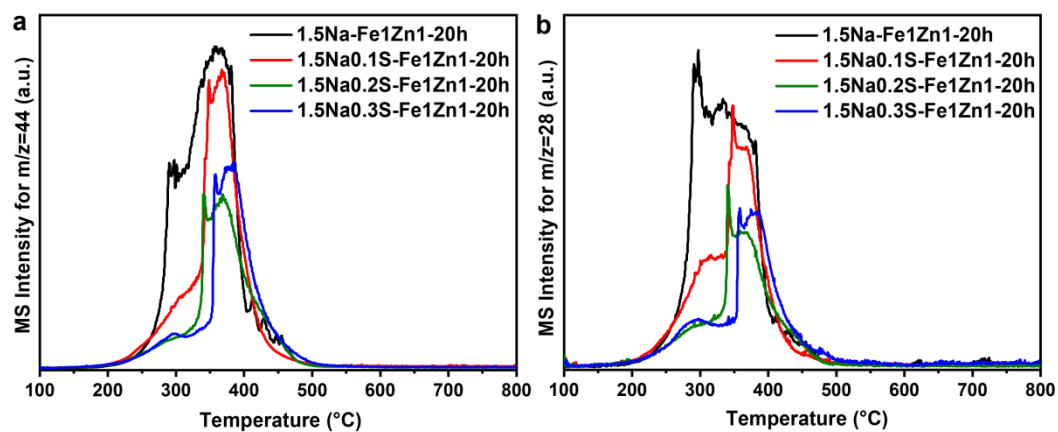


Figure S12. O₂-TPO profiles of the spent NaS-Fe₁Zn₁ catalysts with different Na₂S content after CO-FTS reaction.

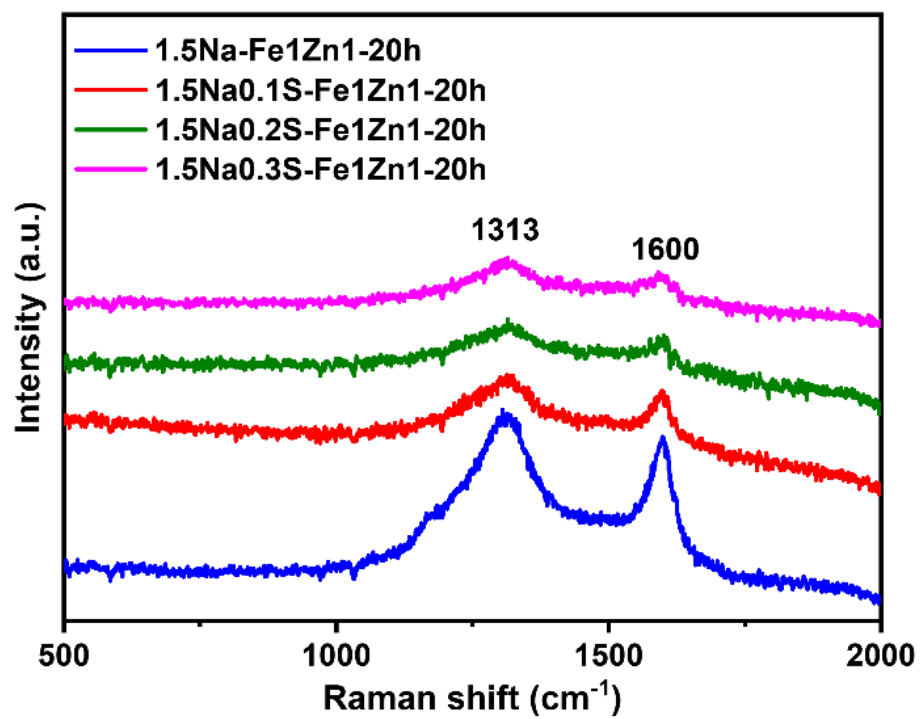


Figure S13. Raman spectra of the spent NaS-Fe1Zn1 catalysts with different Na_2S content after CO-FTS reaction.

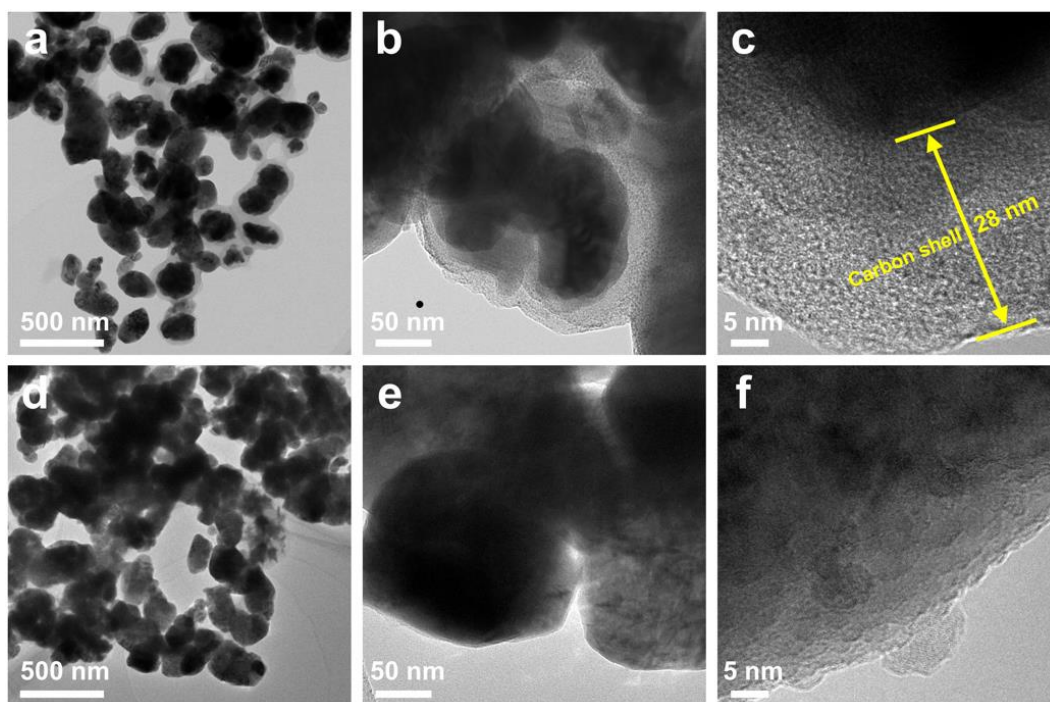


Figure S14. TEM images of spent 1.5Na-Fe1Zn1 (a-c) and 1.5Na0.1S-Fe1Zn1 (d-f) after CO-FTS reaction.

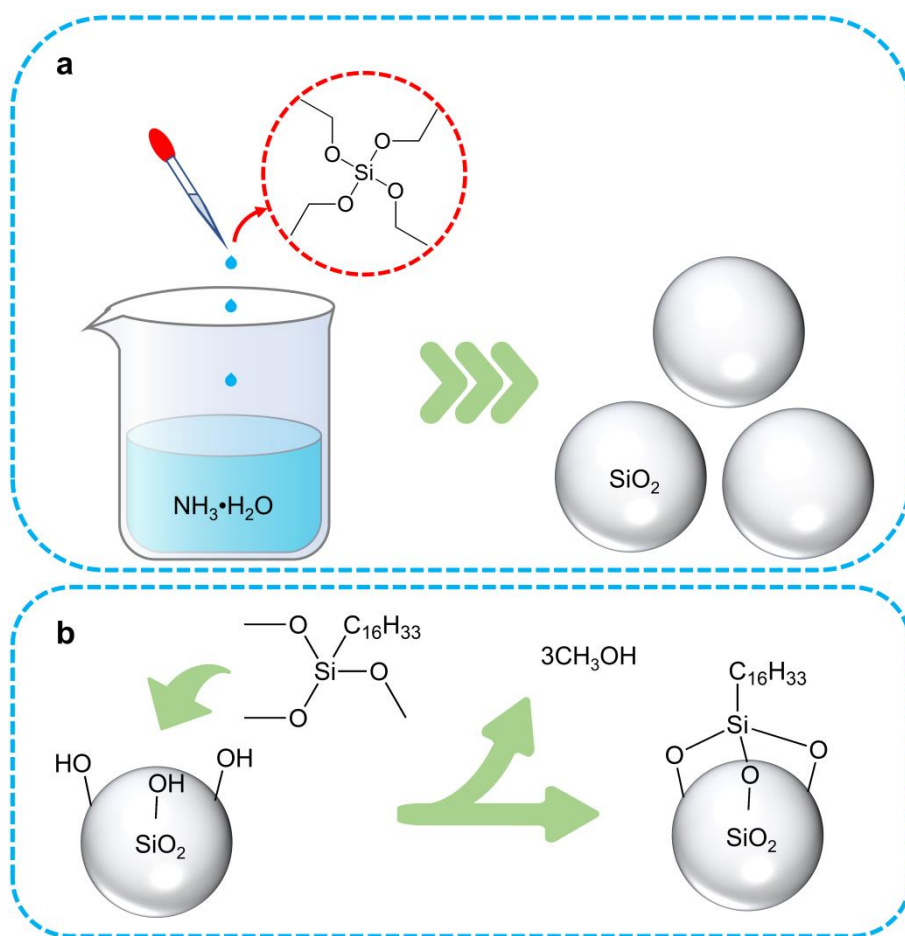


Figure S15. Illustrations of the (a) synthesis of SiO_2 diluter and (b) its hydrophobic modification $\text{SiO}_2@\text{C}_{16}$.

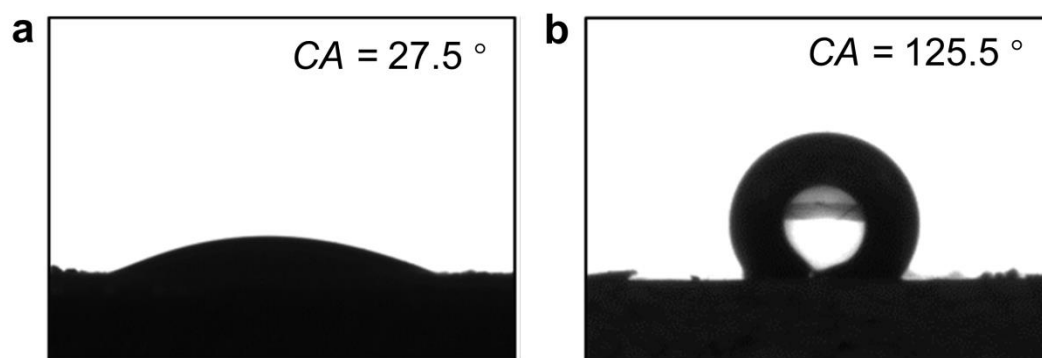


Figure S16. Water contact angle of (a) pure SiO₂ and (b) SiO₂ with hydrophobic modification.

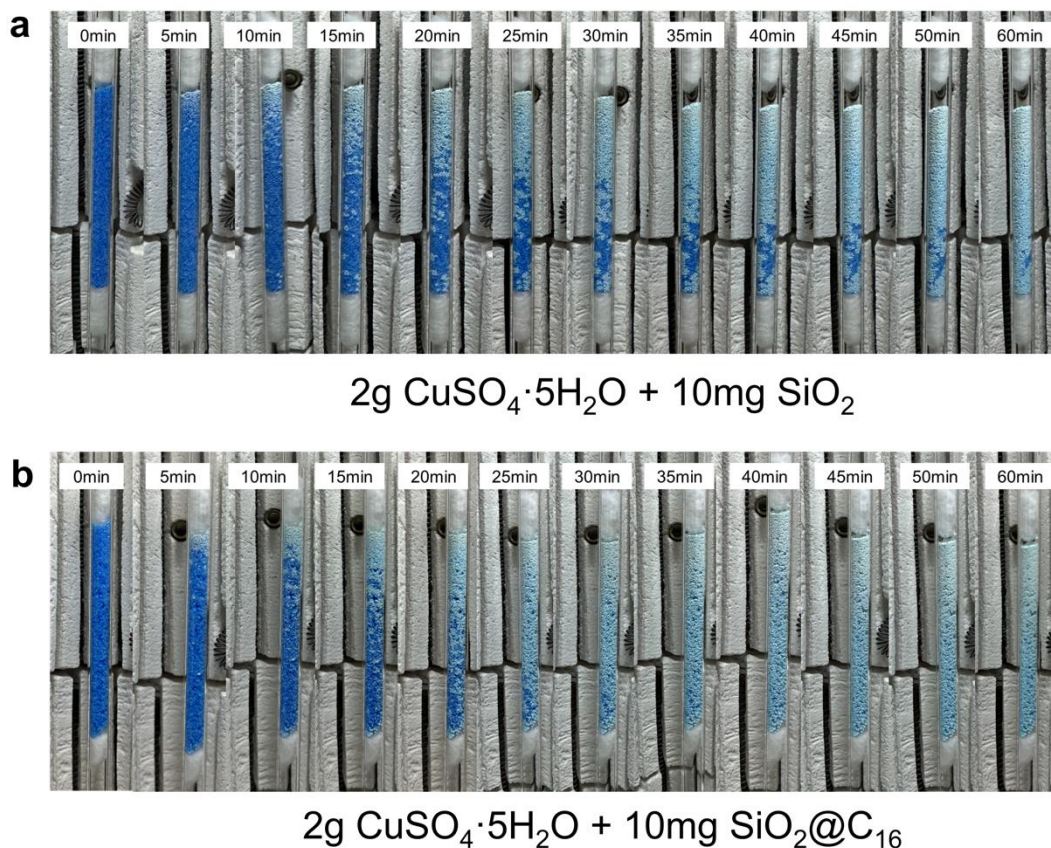


Figure S17. Dehydration experiment of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ mixed with (a) SiO_2 and $\text{SiO}_2 @ \text{C}_{16}$ at 100°C in an Ar flow.

With the mixing of hydrophobic-modified SiO_2 , a faster dehydration rate of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ was achieved. It suggested that in the fixed bed reactor, the mass transfer of H_2O could be affected by the hydrophilicity/hydrophobicity of the diffusion channel. The hydrophobic modification of diffusion channel surface exhibited much weaker interaction with H_2O , favoring the mass transfer of H_2O to leave out of the reactor.

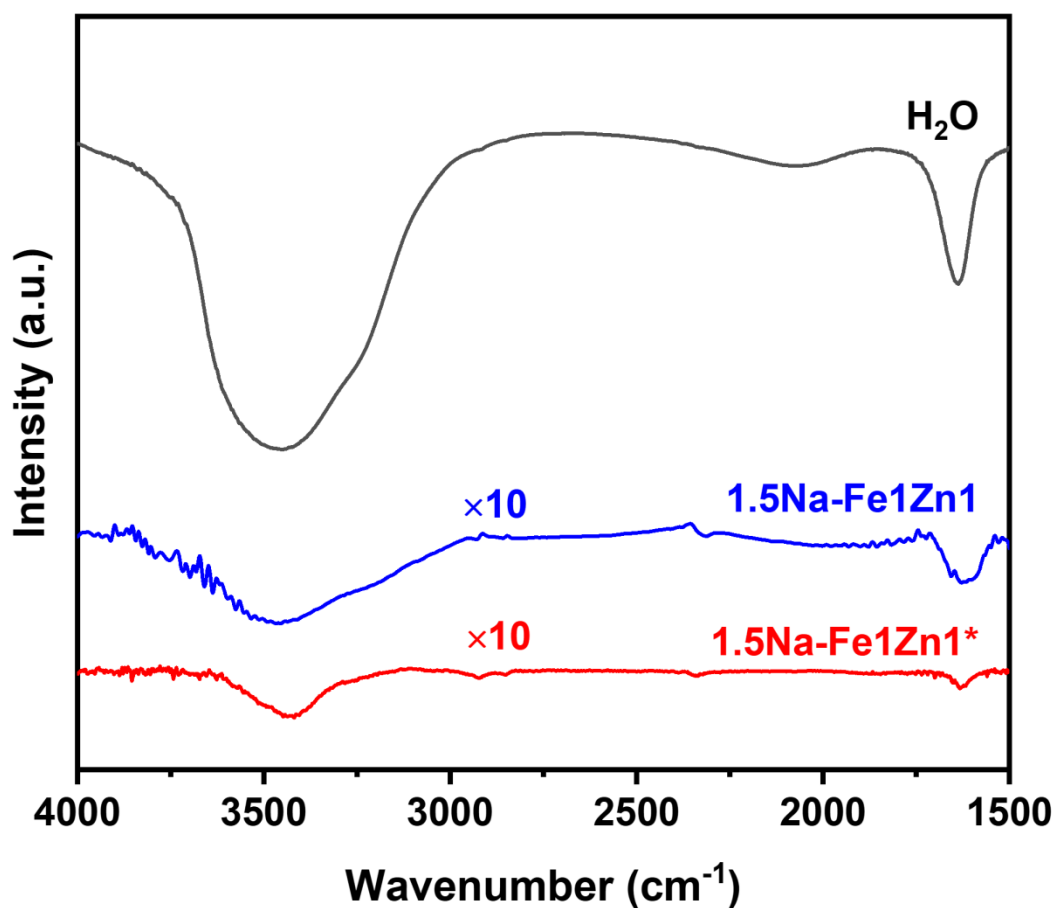


Figure S18. FTIR spectra of the spent 1.5Na-Fe1Zn1 catalyst after atmospheric-pressure CO₂-FTS reaction. Note that: * represented that spent catalyst was collected after CO₂-FTS reaction using hydrophobic treated SiO₂ as diluter.

It can be seen that the spent 1.5Na-Fe1Zn1 catalyst mixed with hydrophobic treated SiO₂ exhibited two lower IR bands of H₂O, suggesting that hydrophobic modification favored to alleviate the accumulation of H₂O around catalyst through enabling a faster mass transfer rate of byproduct H₂O to leave FTS catalyst surface.

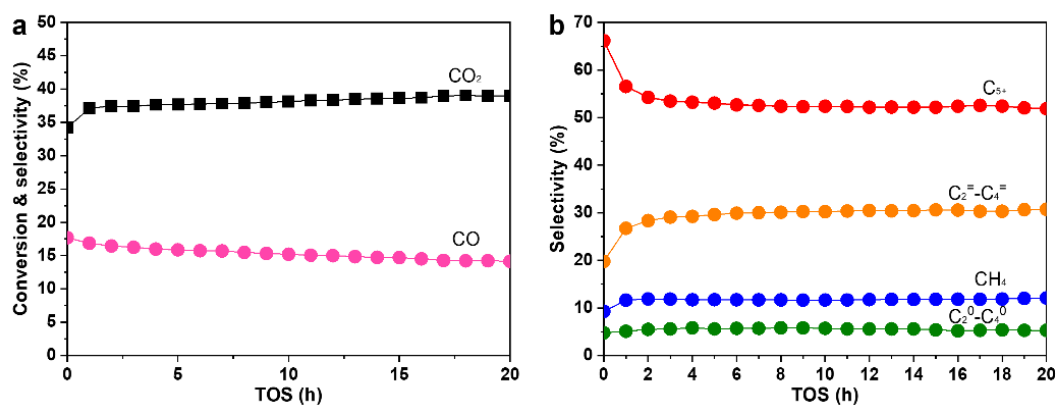


Figure S19. Catalytic performance for the high-temperature CO₂-FTS reaction over 1.5Na-Fe1Zn1+SiO₂@C₁₆ catalyst: (a) CO₂ conversion and CO selectivity, and (b) hydrocarbon product selectivity. Reaction conditions: 2 g catalyst, 320 °C, H₂/CO₂ = 3, 1 MPa, and GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

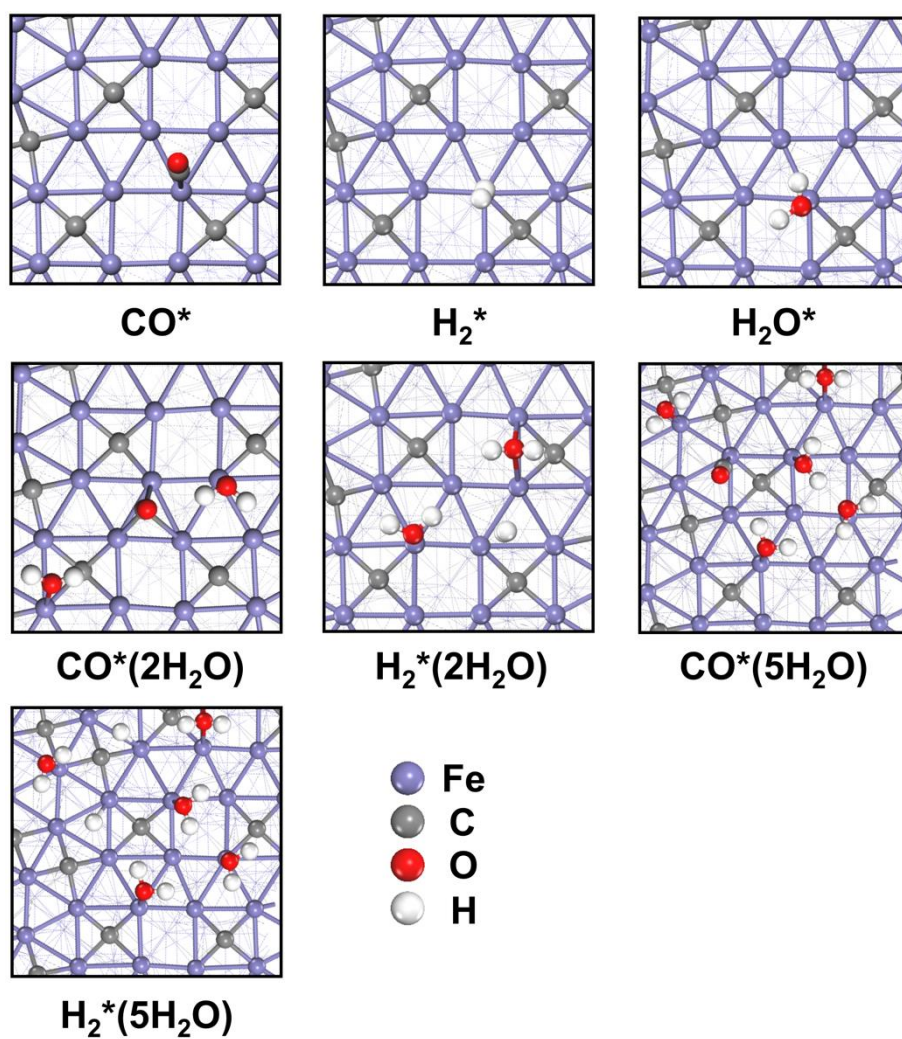


Figure S20. Structures for the adsorption of CO, H₂, and H₂O on pure Fe₅C₂(510) surface and H₂O-adsorbed Fe₅C₂(510) surface.

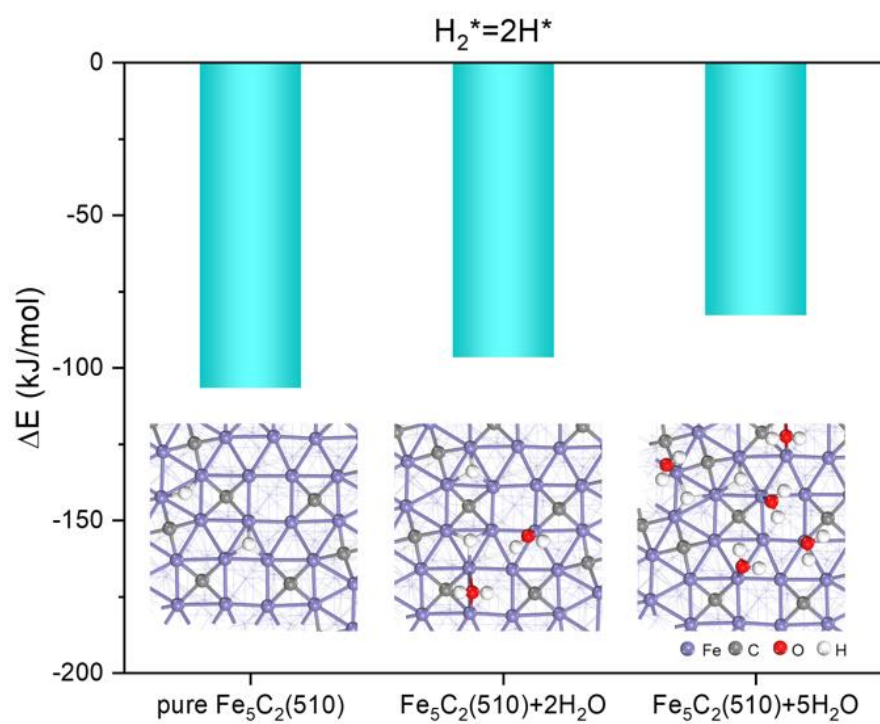


Figure S21. Dissociation energy of H_2 on pure $\text{Fe}_5\text{C}_2(510)$ surface and H_2O -adsorbed $\text{Fe}_5\text{C}_2(510)$ surface.

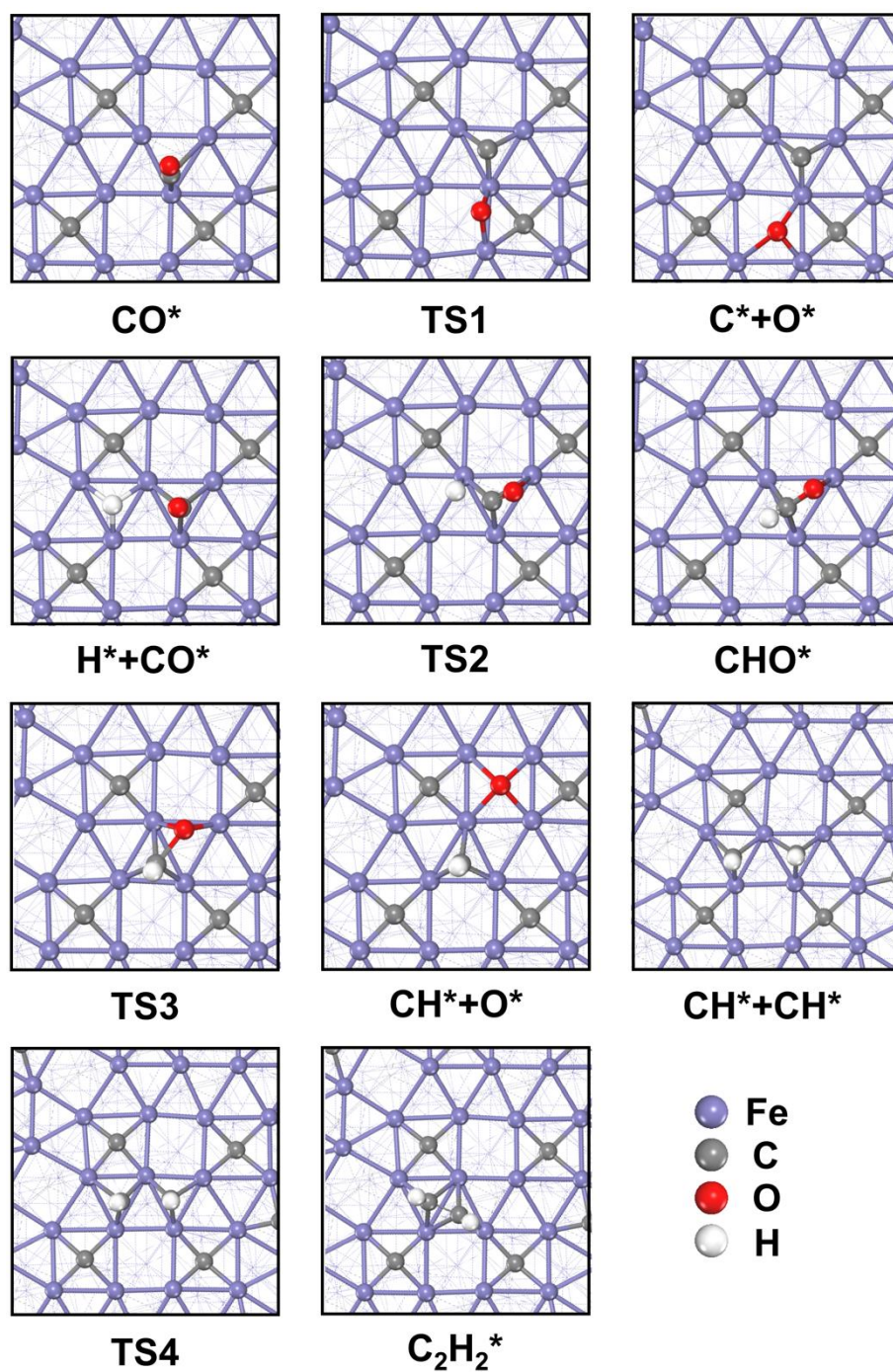


Figure S22. Structures for CO dissociation and C-C coupling on pure $\text{Fe}_5\text{C}_2(510)$ surface.

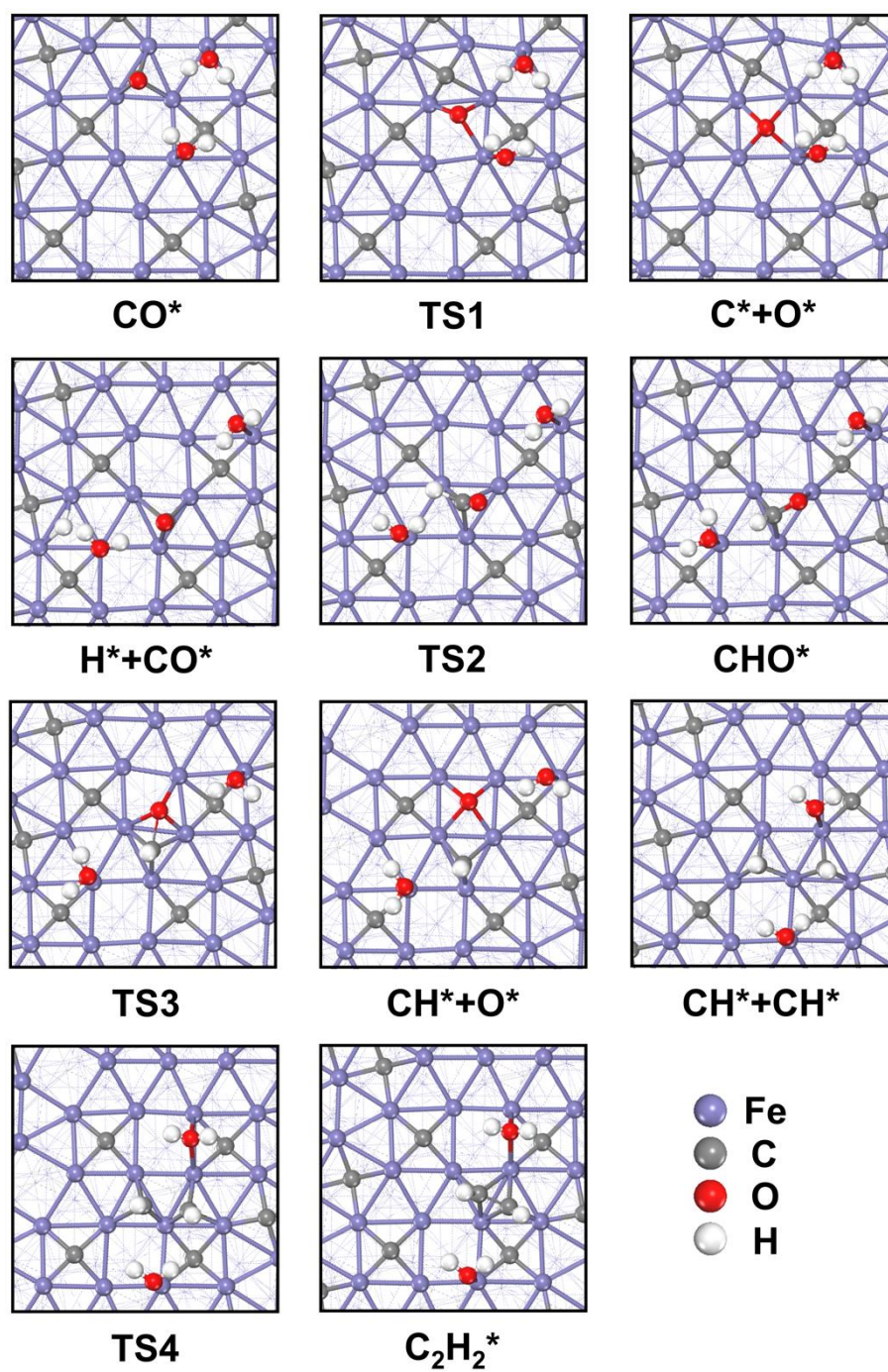


Figure S23. Structures for CO dissociation and C-C coupling on $\text{Fe}_5\text{C}_2(510)$ surface adsorbed with two H_2O molecules.

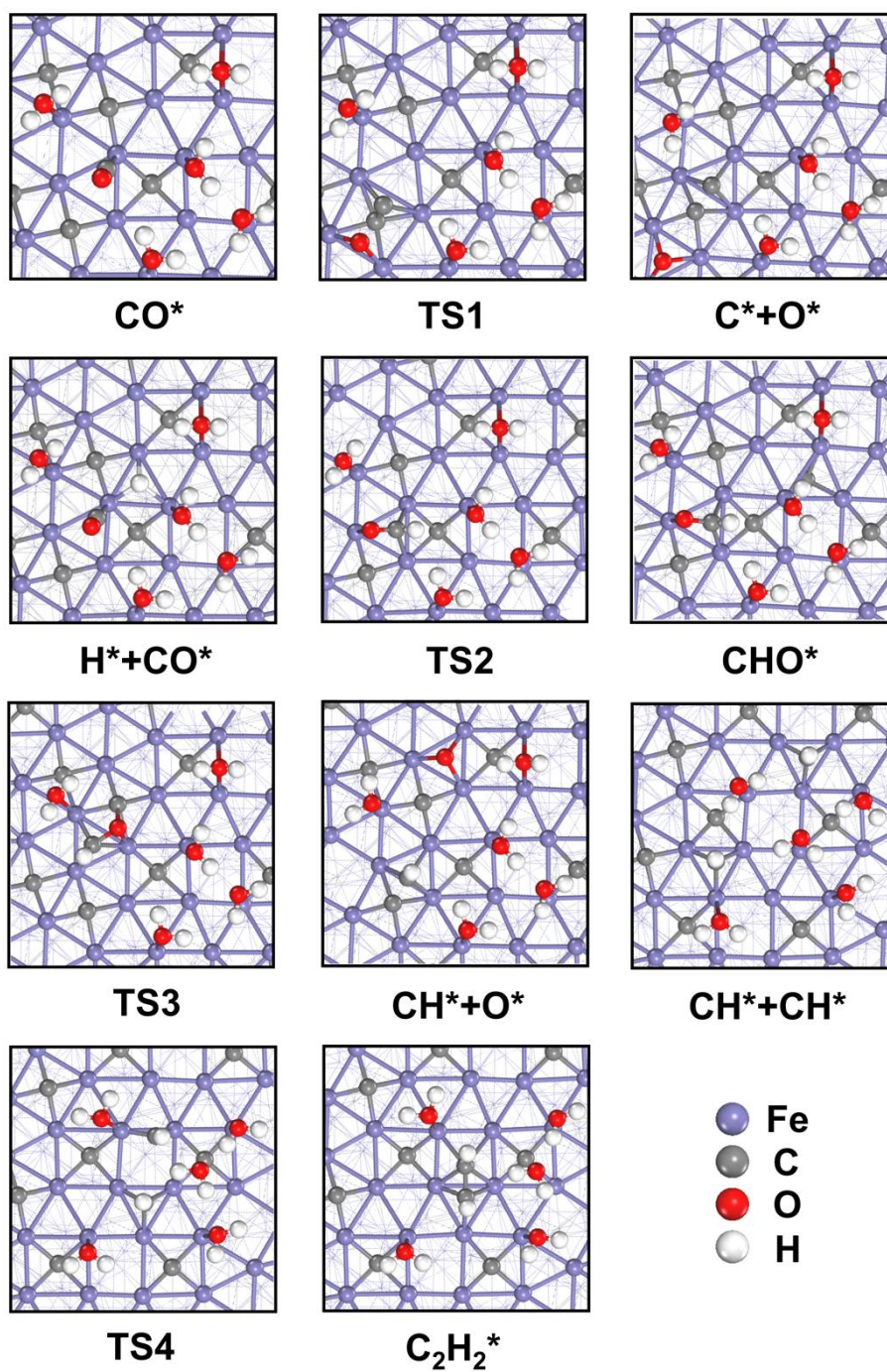


Figure S24. Structures for CO dissociation and C-C coupling on $\text{Fe}_5\text{C}_2(510)$ surface adsorbed with five H_2O molecules.

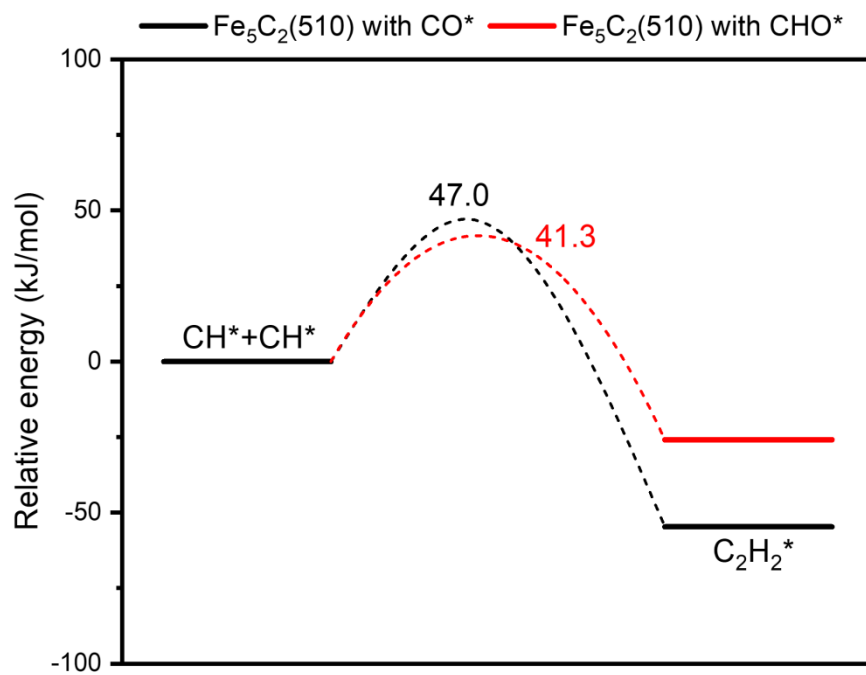


Figure S25. The energy profiles for C-C coupling on CO^* and CHO^* adsorbed $\text{Fe}_5\text{C}_2(510)$ surface.

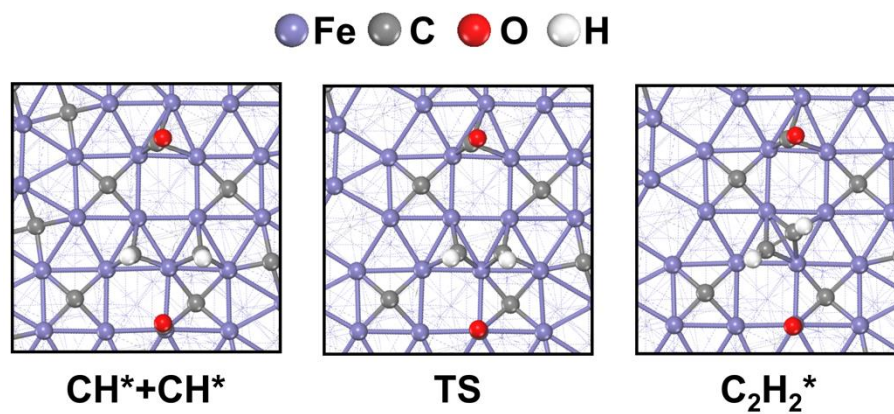


Figure S26. Structures for C-C coupling on CO* adsorbed Fe₅C₂(510) surface.

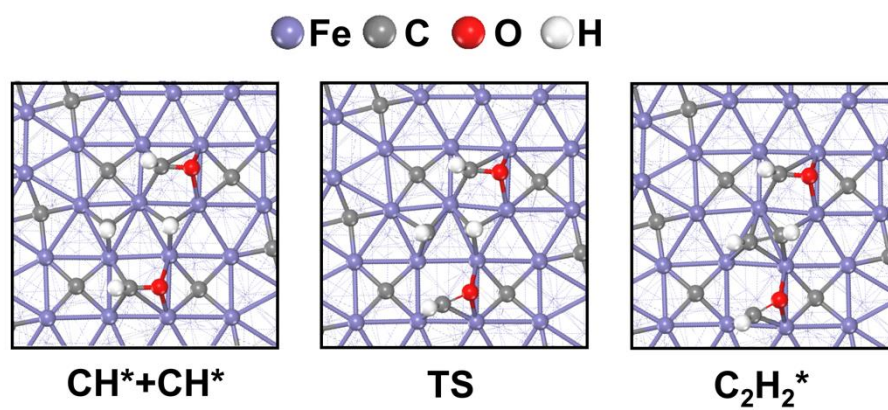


Figure S27. Structures for C-C coupling on CHO* adsorbed Fe₅C₂(510) surface.

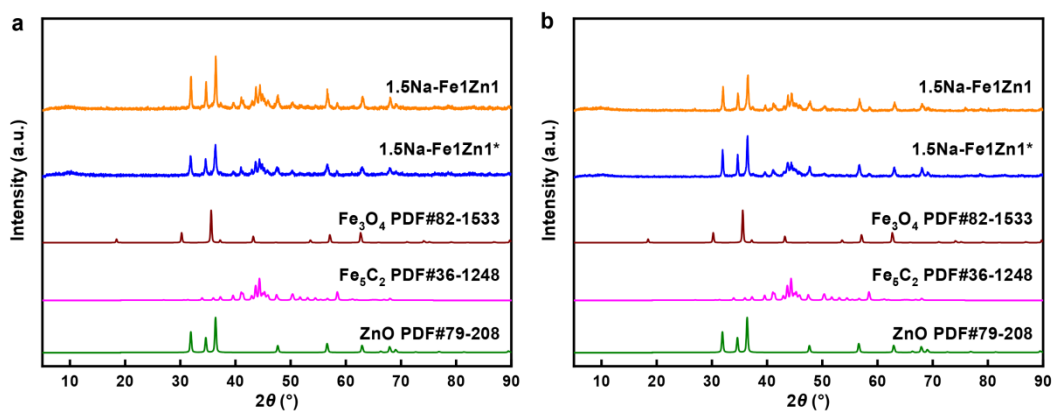


Figure S28. XRD patterns of the spent 1.5Na-Fe1Zn1 catalyst after (a) atmospheric-pressure and (b) high-pressure CO_2 -FTS reaction. Note that: * represented the spent catalyst was collected after CO_2 -FTS reaction using hydrophobic treated SiO_2 as diluter.

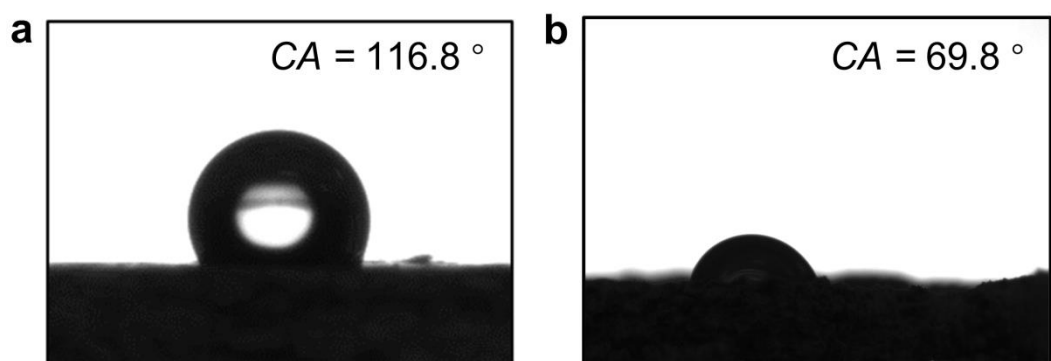


Figure S29. Water contact angle of the SiO₂ with hydrophobic modification after (a) atmospheric-pressure and (b) high-pressure CO₂-FTS reaction.

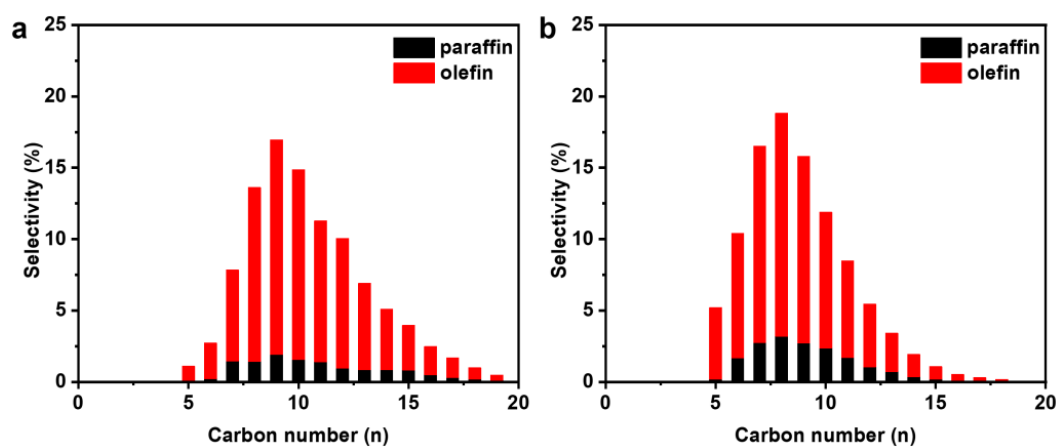


Figure S30. Carbon number distribution of the total liquid-phase products collected during (a) 20 h CO₂-FTS reaction and (b) 1000 h CO₂-FTS reaction over 1.5Na-Fe1Zn1 catalyst.

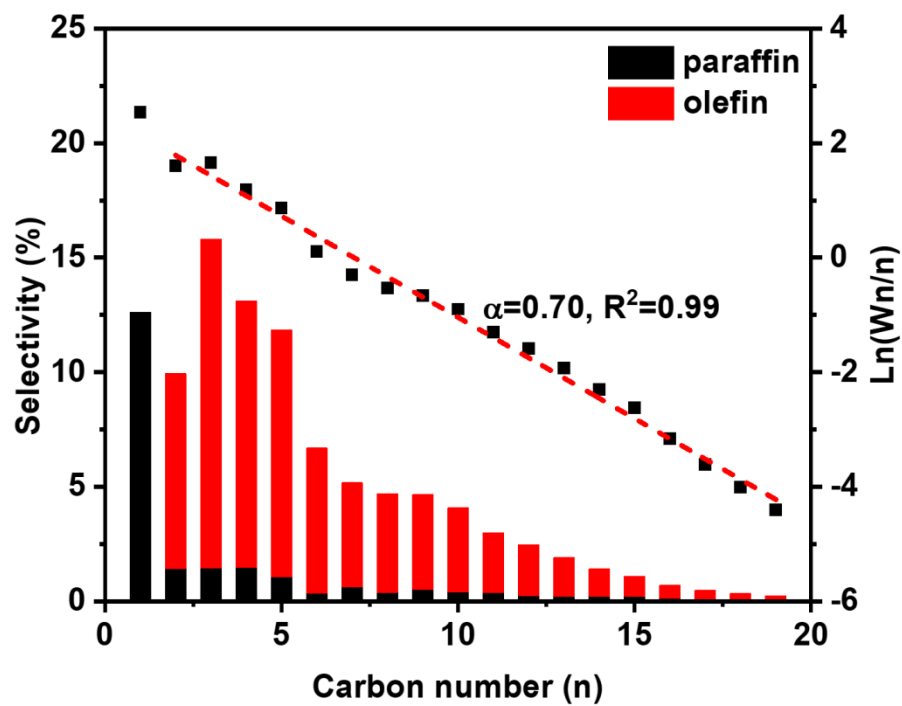


Figure S31. Carbon number distribution of the total products for 20 h CO₂-FTS reaction over 1.5Na-Fe1Zn1 catalyst.

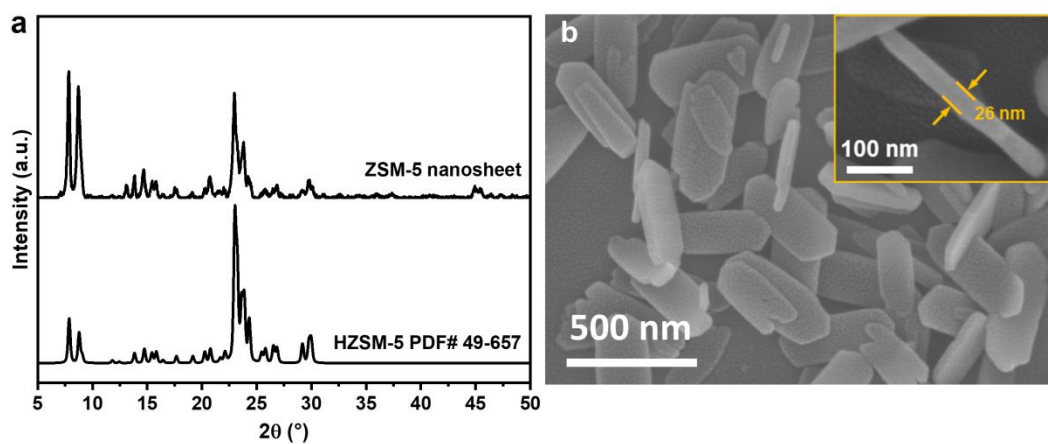


Figure S32. Characterizations of ZSM-5 nanosheet. (a) XRD pattern, and (b) SEM image.

As shown in Figure S32a, as-synthesized ZSM-5 nanosheet catalyst exhibited typical MFI-structure XRD peaks (PDF# 49-657). As shown in Figure S32b, as-synthesized ZSM-5 showed a plate-like morphology with the thickness of about 30 nm.

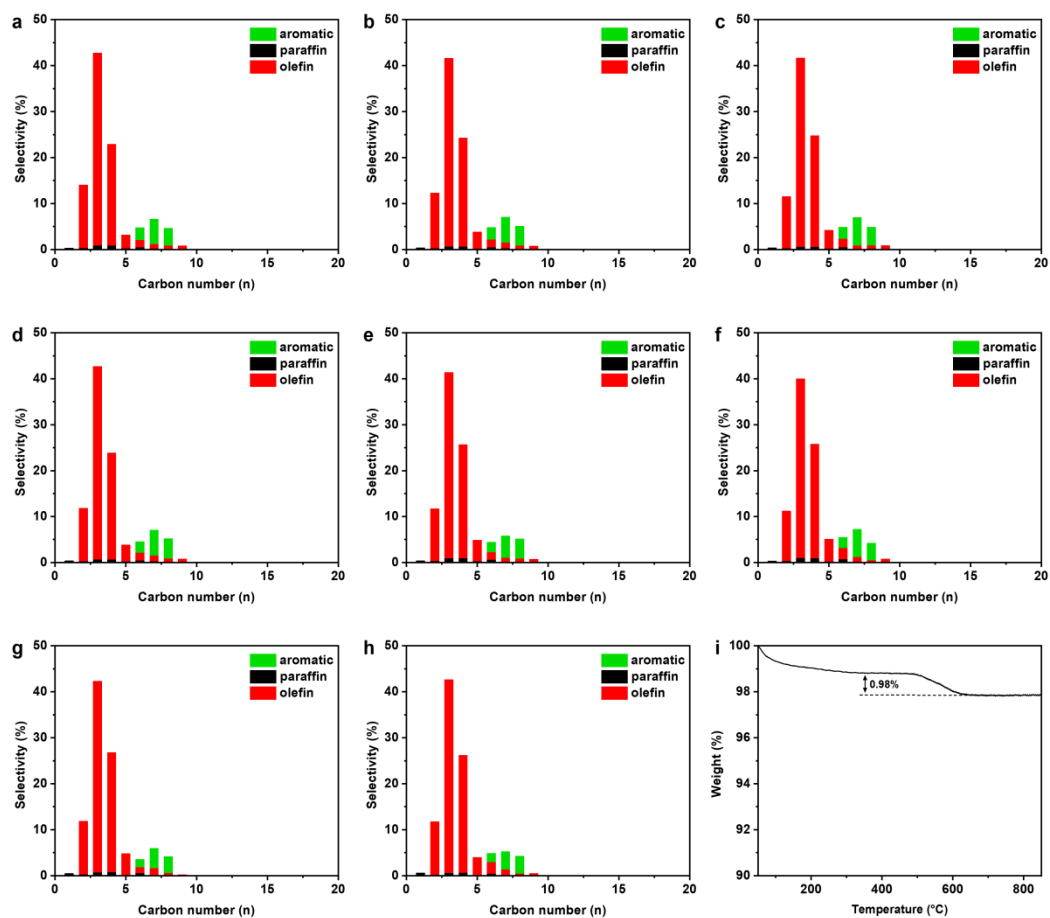


Figure S33. Catalytic cracking of collected CO₂-FTS liquid phase products. (a-h) carbon number distribution of cracking products. Time on stream is 0.3 h, 1.7 h, 3 h, 4.3 h, 5.7 h, 7 h, 8.3h and 9.7 h for (a-h), respectively. (i) TG profiles of spent catalyst after 10 h catalytic test.

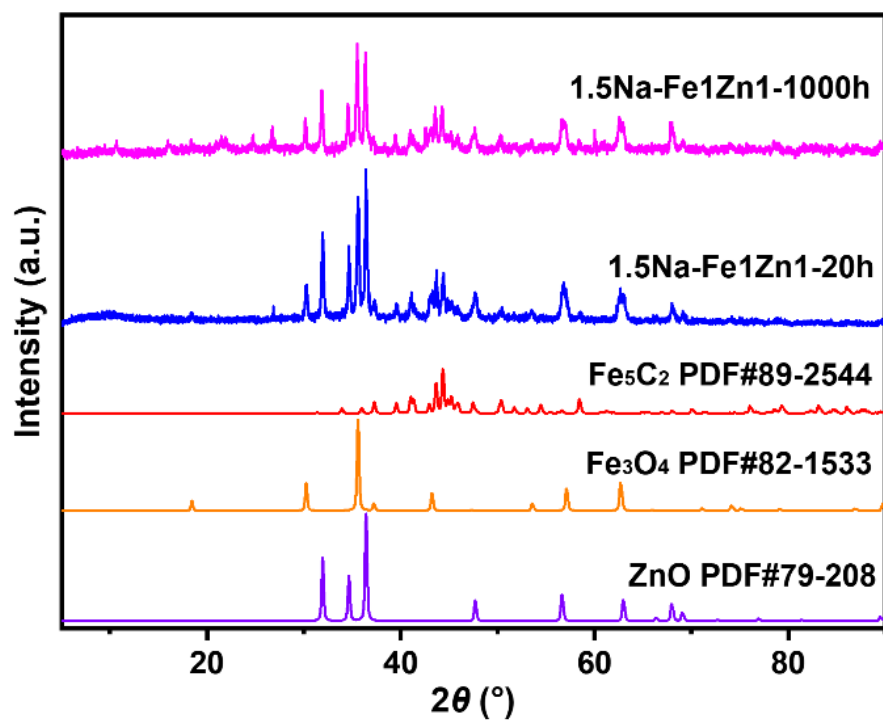


Figure S34. XRD profiles of the spent 1.5Na-Fe1Zn1 catalyst after 20 h and 1000 h CO₂-FTS reaction.

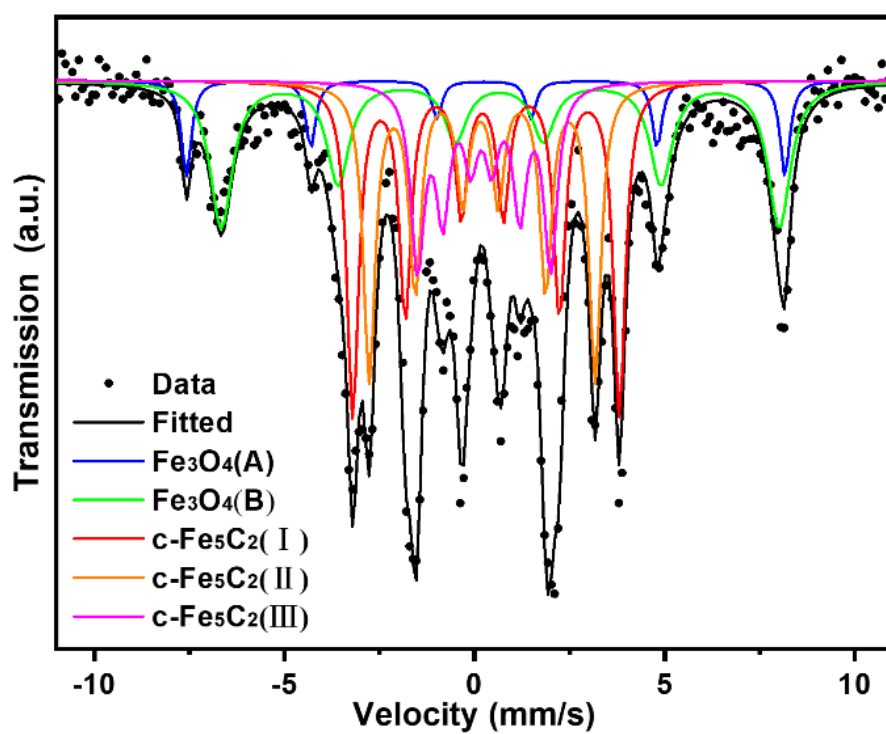


Figure S35. Mössbauer spectrum of the spent 1.5Na-Fe1Zn1 catalyst after 1000 h CO₂-FTS reaction.

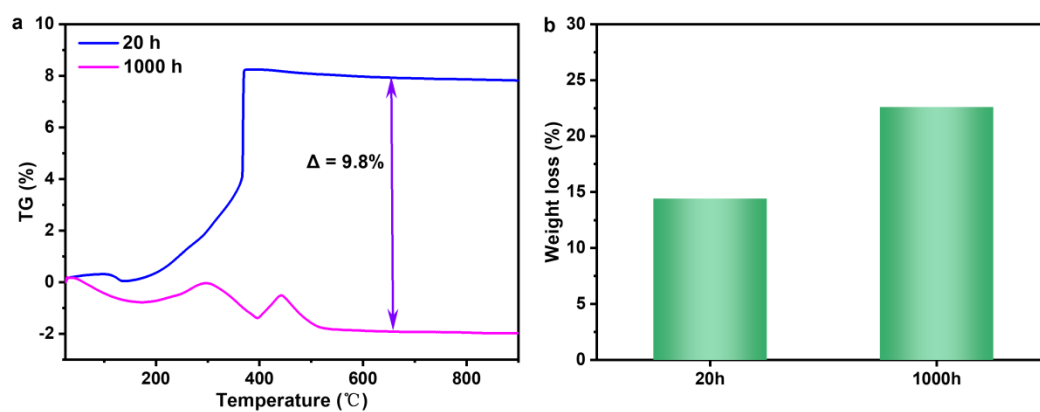


Figure S36. (a)TG profiles and (b) calculated weight loss of the spent 1.5Na-Fe1Zn1 catalyst after 20 h and 1000 h CO₂-FTS reaction.

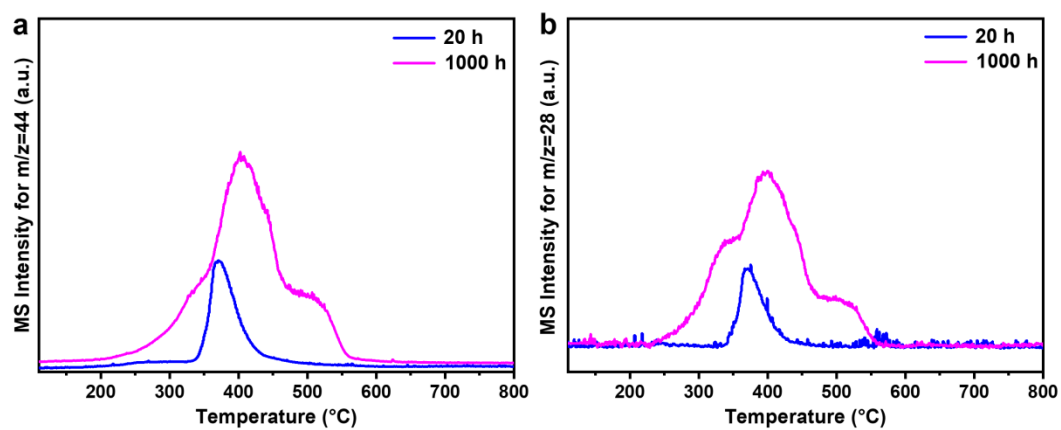


Figure S37. O₂-TPO profiles of the spent 1.5Na-Fe₁Zn₁ catalyst after 20 h and 1000 h CO₂-FTS reaction: (a) CO₂ signal, and (b) CO signal.

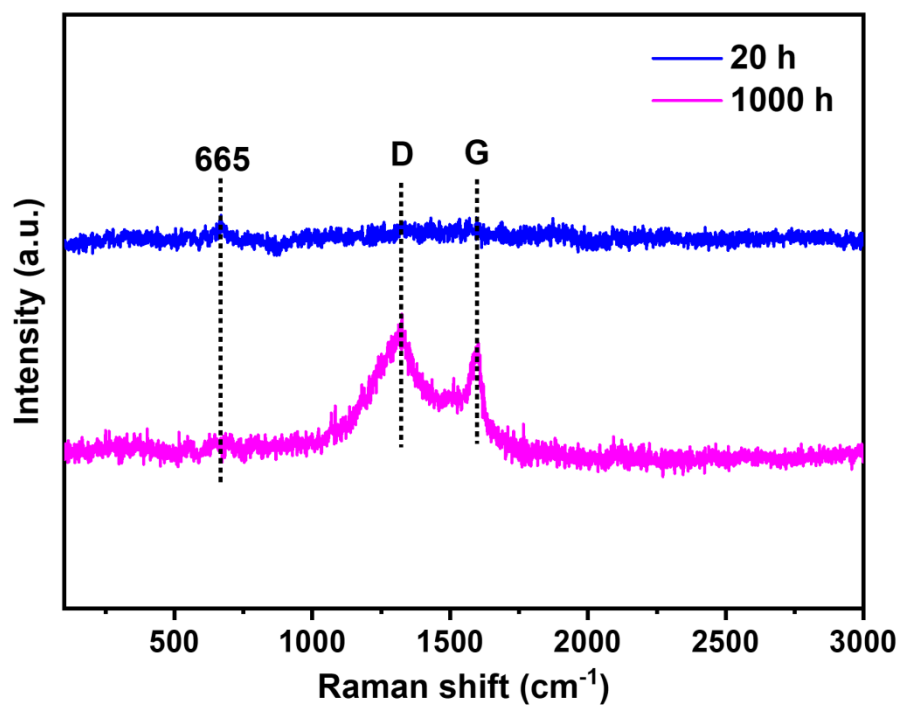


Figure S38. Raman spectra of the spent 1.5Na-Fe1Zn1 catalyst after 20h and 1000 h CO₂-FTS reaction.

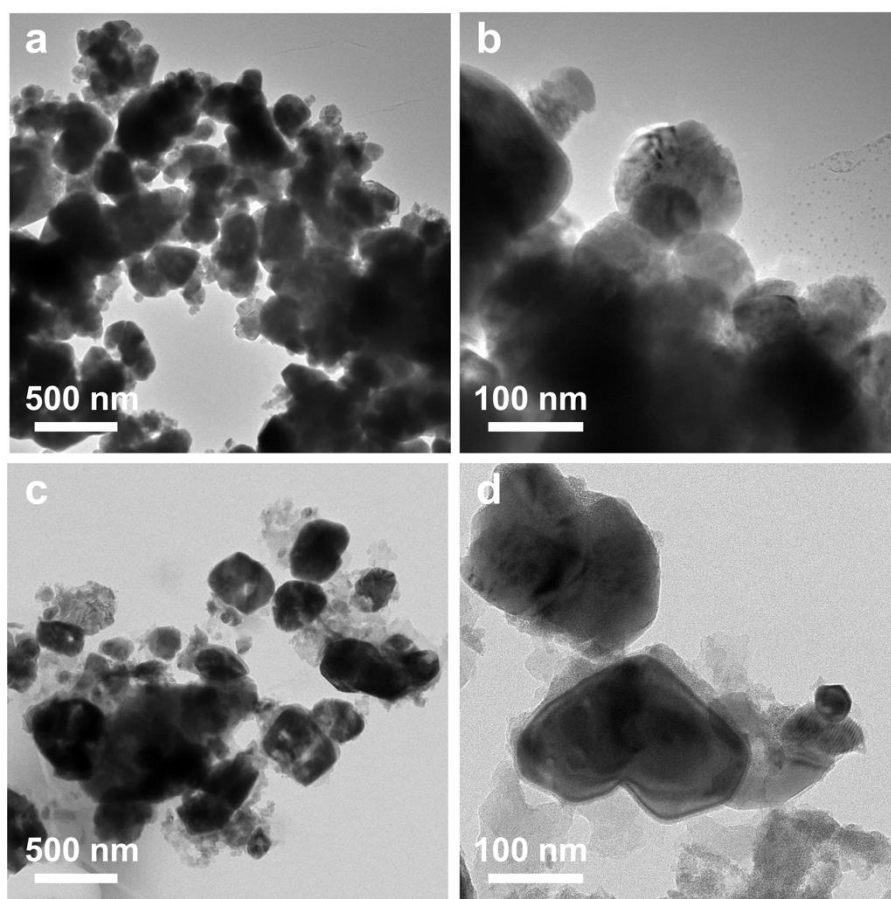


Figure S39. TEM images of the spent 1.5Na-Fe1Zn1 catalyst after (a, b) 20 h and (c, d) 1000 h CO₂-FTS reaction.

Table S1. Comparison of the catalytic performance in the last five hours for CO₂-FTS reaction over 1.5Na-FeZn catalysts with different Fe/Zn ratio.

Catalyst	X _{CO2} (%)	S _{CO} (%)	Hydrocarbon Selectivity (%)				O/(O+P) (%)
			CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ⁰ -C ₄ ⁰	C ₅₊	
1.5Na-Fe	38.5	16.7	10.6	33.4	3.9	52.0	89.4
1.5Na-Fe4Zn1	38.3	16.3	13.1	32.5	4.4	50.0	88.1
1.5Na-Fe2Zn1	38.9	15.4	13.0	34.2	4.5	48.3	88.4
1.5Na-Fe1Zn1	39.0	15.6	12.7	34.5	4.3	48.5	88.9

Reaction conditions: 2 g catalyst, 320 °C, H₂/CO₂ = 3, 1 MPa, and GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

Table S2. Comparison of the catalytic performance in the last five hours for CO₂-FTS reaction over Na-Fe1Zn1 catalysts with different Na loading.

Catalyst	X _{CO2} (%)	S _{CO} (%)	Hydrocarbon Selectivity (%)				O/(O+P) (%)
			CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ⁰ -C ₄ ⁰	C ₅₊	
Fe1Zn1	28.8	21.1	48.7	1.2	44.4	5.7	21.1
0.75Na-Fe1Zn1	38.7	16.4	12.5	34.5	4.2	48.8	89.1
1.5Na-Fe1Zn1	39.0	15.6	12.7	34.5	4.3	48.5	88.9
3Na-Fe1Zn1	39.0	14.8	11.8	32.9	4.1	51.2	88.9

Reaction conditions: 2 g catalyst, 320 °C, H₂/CO₂ = 3, 1 MPa, and GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

Table S3. Comparison of the catalytic performance in the last five hours for CO₂-FTS reaction over 1.5Na-Fe1Zn1 catalyst under different reaction temperature.

Temperature	X _{CO2} (%)	S _{CO} (%)	Hydrocarbon Selectivity (%)				O/(O+P) (%)
			CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ⁰ -C ₄ ⁰	C ₅₊	
280 °C	25.2	22.6	10.7	34.7	6.0	48.6	85.3
300 °C	33.0	16.4	10.0	32.2	4.0	53.7	88.9
320 °C	39.0	15.6	12.7	34.5	4.3	48.5	88.9
340 °C	42.8	14.9	15.9	29.6	3.8	50.6	88.6

Reaction conditions: 2 g catalyst, H₂/CO₂ = 3, 1 MPa, and GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

Table S4. Comparison of the catalytic performance in the last five hours for CO₂-FTS reaction over 1.5Na-Fe1Zn1 catalyst under different reaction pressure.

Pressure	X _{CO2} (%)	S _{CO} (%)	Hydrocarbon Selectivity (%)				O/(O+P) (%)
			CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ⁰ -C ₄ ⁰	C ₅₊	
0.5 MPa	35.9	21.6	15.9	36.2	4.4	43.6	89.4
1.0 MPa	39.0	15.6	12.7	34.5	4.3	48.5	88.9
1.5 MPa	39.9	13.7	12.8	34.2	4.7	48.3	87.9
2.0 MPa	40.5	12.5	13.5	33.9	5.0	47.6	87.2

Reaction conditions: 2 g catalyst, 320 °C, H₂/CO₂ = 3, and GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

Table S5. Comparison of the catalytic performance in the last five hours for CO₂-FTS reaction over 1.5Na-Fe1Zn1 catalyst under different reaction GHSV.

GHSV (mL g _{cat} ⁻¹ h ⁻¹)	X _{CO2} (%)	S _{CO} (%)	Hydrocarbon Selectivity (%)				O/(O+P) (%)
			CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ⁰ -C ₄ ⁰	C ₅₊	
600	39.7	14.3	12.5	29.8	4.4	53.3	87.0
1200	39.0	15.6	12.7	34.5	4.3	48.5	88.9
2400	37.8	15.8	12.4	32.1	3.8	51.7	89.4
4800	34.8	19.1	14.7	34.4	4.3	46.6	88.8

Reaction conditions: 2 g catalyst, 320 °C, 1 MPa, and H₂/CO₂ = 3.

Table S6. Comparison of the catalytic performance in the last five hours for CO₂-FTS reaction over 1.5Na-Fe1Zn1 catalyst under different H₂/CO₂.

H ₂ /CO ₂	X _{CO2} (%)	S _{CO} (%)	Hydrocarbon Selectivity (%)				O/(O+P) (%)
			CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ⁰ -C ₄ ⁰	C ₅₊	
1	17.5	30.5	11.6	29.5	3.7	55.2	88.9
2	28.9	20.7	11.9	31.0	4.7	52.4	86.8
3	39.0	15.6	12.7	34.5	4.3	48.5	88.9

Reaction conditions: 2 g catalyst, 320 °C, 1 MPa, and GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

Table S7. Comparison of the catalytic performance of CO₂-FTS in the last five hours over NaS-Fe1Zn1 catalysts with different Na₂S content.

Catalyst	X _{CO2} (%)	S _{CO} (%)	Hydrocarbon Selectivity (%)				O/(O+P) (%)
			CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ⁰ -C ₄ ⁰	C ₅₊	
1.5Na-Fe1Zn1	39.0	15.6	12.7	34.5	4.3	48.5	88.9
1.5Na0.1S-Fe1Zn1	24.8	42.6	55.9	4.2	38.5	1.4	9.8
1.5Na0.2S-Fe1Zn1	15.1	71.3	58.9	5.2	33.8	2.1	13.3

Reaction conditions: 2.0 g catalyst, 320 °C, H₂/CO₂ = 3, 1 MPa, and GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

Table S8. Textural properties and promoter contents of the fresh NaS-Fe1Zn1 catalysts.

Catalysts	Specific surface area (m ² g ⁻¹)	Pore volume (cm ³ g ⁻¹)	Average pore diameter (nm)	Na (wt%)	S (wt%)
1.5Na-Fe	37	0.74	5.90	1.72	---
1.5Na-Fe1Zn1	47	0.49	3.10	1.79	---
1.5Na0.1S-Fe1Zn1	44	0.56	3.80	1.75	0.11
1.5Na0.2S-Fe1Zn1	45	0.51	3.30	1.79	0.22

Table S9. Iron phase composition of the spent 1.5Na-Fe1Zn1 and 1.5Na0.2S-Fe1Zn1 catalyst after 20 h CO₂-FTS reaction from Mössbauer spectroscopy.

Catalysts	Phases	Mössbauer parameters				Area (%)
		Hhf (kOe)	IS (mm/s)	QS (mm/s)	Γ/2 (mm/s)	
1.5Na-Fe1Zn1-20h	Fe ₃ O ₄ (A)	490.82	0.27	-0.01	0.12	7.5
	Fe ₃ O ₄ (B)	455.74	0.65	0.01	0.33	24.5
	χ-Fe ₅ C ₂ (I)	215.38	0.25	-0.10	0.19	29.3
	χ-Fe ₅ C ₂ (II)	184.63	0.19	-0.05	0.20	23.0
	χ-Fe ₅ C ₂ (III)	107.98	0.22	0.05	0.22	15.7
1.5Na0.2S-Fe1Zn1-20h	Fe ⁰	330.62	-0.01	-0.02	0.13	30.7
	χ-Fe ₅ C ₂ (I)	215.73	0.22	-0.09	0.14	17.9
	Fe ₃ C	204.45	0.20	-0.04	0.17	28.0
	χ-Fe ₅ C ₂ (II)	185.61	0.22	-0.01	0.21	13.9
	χ-Fe ₅ C ₂ (III)	112.77	0.21	0.04	0.20	9.5

Table S10. Comparison of the catalytic performance in the last five hours for CO-FTS reaction over NaS-Fe1Zn1 catalysts with different Na₂S content.

Catalyst	X _{CO} (%)	S _{CO2} (%)	Hydrocarbon Selectivity (%)				O/(O+P) (%)
			CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ⁰ -C ₄ ⁰	C ₅₊	
1.5Na-Fe1Zn1	13.9	41.1	35.6	11.0	3.9	49.6	73.6
1.5Na0.1S-Fe1Zn1	96.9	35.9	5.8	22.5	3.9	67.8	85.1
1.5Na0.2S-Fe1Zn1	95.6	36.8	5.6	26.4	4.2	63.8	86.3
1.5Na0.3S-Fe1Zn1	89.4	38.8	5.6	27.8	3.7	62.9	88.1

Reaction conditions: 2.0 g catalyst, 320 °C, H₂/CO = 1, 1 MPa, and GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

Table S11 Comparison of the catalytic performance in the last five hours for CO-FTS reaction over 1.5Na0.1S-Fe1Zn1 catalyst under different reaction temperature.

Temperature	X_{CO} (%)	S_{CO_2} (%)	Hydrocarbon Selectivity (%)				O/(O+P)
			CH_4	$C_2^=-C_4^=-$	$C_2^0-C_4^0$	C_{5+}	
280 °C	24.0	39.8	4.8	29.2	2.5	63.5	92.1
300 °C	83.7	39.3	5.3	27.4	4.0	63.3	87.3
320 °C	96.9	35.9	5.8	22.5	3.9	67.8	85.1

Reaction conditions: 2.0 g of catalyst, $H_2/CO = 1$, 1 MPa, GHSV = 1200 mL $g_{cat}^{-1} h^{-1}$.

Table S12 Comparison of the catalytic performance in the last five hours for CO-FTS reaction over 1.5Na0.1S-Fe1Zn1 catalyst under different H₂/CO.

H ₂ /CO	X _{CO} (%)	S _{CO2} (%)	Hydrocarbon Selectivity (%)				O/(O+P)
			CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ⁰ -C ₄ ⁰	C ₅₊	
1	96.9	35.9	5.8	22.5	3.9	67.8	85.1
2	95.6	31.3	9.5	31.0	6.7	52.9	82.2
3	94.9	26.8	12.2	30.4	10.9	46.5	73.7

Reaction conditions: 2.0 g of catalyst, 320 °C, 1 MPa, GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

Table S13 Comparison of the catalytic performance in the last five hours for CO-FTS reaction over 1.5Na0.1S-Fe1Zn1 catalyst under different reaction pressure.

Pressure	X_{CO} (%)	S_{CO_2} (%)	Hydrocarbon Selectivity (%)				O/(O+P)
			CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ⁰ -C ₄ ⁰	C ₅₊	
0.5MPa	88.0	38.3	4.9	24.5	3.5	67.2	87.6
1.0MPa	96.9	35.9	5.8	22.5	3.9	67.8	85.1
1.5MPa	97.5	34.3	5.7	20.0	3.8	70.5	84.2

Reaction conditions: 2.0 g of catalyst, 320 °C, H₂/CO = 1, GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

Table S14. Iron phase composition of the spent catalysts after CO-FTS from Mössbauer spectroscopy.

Catalysts	Phases	Mössbauer parameters				Area (%)
		Hhf (kOe)	IS (mm/s)	QS (mm/s)	$\Gamma/2$ (mm/s)	
1.5Na-Fe1Zn1-20h	Fe ²⁺ /Fe ³⁺	—	1.29	0.50	0.18	2.4
	χ -Fe ₅ C ₂ (I)	216.77	0.24	-0.09	0.17	39.2
	Fe ₃ C	196.29	0.19	-0.04	0.11	12.1
	χ -Fe ₅ C ₂ (II)	181.00	0.17	-0.05	0.17	28.8
	χ -Fe ₅ C ₂ (III)	109.67	0.23	-0.12	0.16	17.5
1.5Na0.1S-Fe1Zn1-20h	Fe ²⁺ /Fe ³⁺	—	1.08	0.82	0.35	4.6
	χ -Fe ₅ C ₂ (I)	216.77	0.25	-0.09	0.17	38.2
	Fe ₃ C	196.28	0.19	-0.05	0.11	10.4
	χ -Fe ₅ C ₂ (II)	181.45	0.17	-0.06	0.17	27.5
	χ -Fe ₅ C ₂ (III)	109.12	0.23	-0.14	0.17	19.3

Reaction conditions: 2.0 g catalyst, 320 °C, H₂/CO = 1, 1 MPa, and GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

Table S15. Comparison of the catalytic performance in the last five hours for CO₂-FTS reaction over 1.5Na-Fe1Zn1 catalyst with and without hydrophobic modification.

Catalyst	X _{CO2} (%)	S _{CO} (%)	Hydrocarbon Selectivity (%)				O/(O+P) (%)
			CH ₄	C ₂ ⁼ -C ₄ ⁼	C ₂ ⁰ -C ₄ ⁰	C ₅ +	
1.5Na-Fe1Zn1	39.0	15.6	12.7	34.5	4.3	48.5	88.9
1.5Na-Fe1Zn1+SiO ₂ @C ₁₆	38.0	15.3	11.7	29.4	5.5	53.4	84.2

Reaction conditions: 2 g catalyst, 320 °C, H₂/CO₂ = 3, 1 MPa, and GHSV = 1200 mL g_{cat}⁻¹ h⁻¹.

Table S16. Comparison of the catalytic performance for CO₂ hydrogenation to lower olefins through zeolite-contained modified-FTS and methanol-mediated routes.

Catalysts	Temp. (°C)	Press. (MPa)	GHSV (mL g ⁻¹ h ⁻¹)	TOS (h)	X _{CO2} (%)	S _{CO} (%)	Product distribution (%)				Ref.
							CH ₄	C ₂ ⁰ -C ₄ ⁰	C ₂ ⁼ -C ₄ ⁼	C ₅ +	
1.5Na-Fe1Zn1*+ ZSM-5**	320*; 550**	1*; 0.1**	1200*; 4**	1000*; 10**	38.7	14.5	16.7	5.8	55.2	22.3	This work
Fe ₂ O ₃ @KO ₂ /ZSM-5	375	3	5000	NA	48.9	NA	13.9	NA	12.1	NA	[1]
Fe ₂ O ₃ @KO ₂ /MOR	375	3	NA	NA	48.3	NA	14.5	NA	33.3	NA	[1]
Fe/K-6Ca/ZSM-5	375	3	10000	NA	47.0	8	15.0	NA	38.0	25	[2]
Fe ₂ O ₃ @KO ₂ /ZrS/SAPO-34	375	3	5000	50	47.0	13	23	2	50.0	25	[3]
NiCu/CeO ₂ -SAPO-34	375	2	12000	10	17.0	64.2	2.1	16.7	78.0	3.2	[4]
In ₂ O ₃ /SAPO-34	380	3	9000	10	15.3	68.3	2.7	13.7	81.9	1.7	[5]
ZnZrO/SAPO-34	380	2	3600	50	12.6	47	3.0	14.0	80.0	3	[6]
ZnGa ₂ O ₄ /SAPO-34	370	3	5400	10	13.6	49	NA	8.0	88.0	2	[7]
ZnAl ₂ O ₄ /SAPO-34	370	3	5400	10	15.0	49	0.7	10	87.0	2.2	[8]
CuZnZr/SAPO-34	400	2	3000	24	12.6	56.7	8.8	45.7	42.5	NA	[9]
Cr ₂ O ₃ /SAPO-34	370	0.5	4000	50	13.1	36.0	1.2	NA	95.7	NA	[10]
In-Zr/SAPO-34	400	3	9000	150	35.5	80.0	5.0	~14.6	76.4	~4.0	[11]
In ₂ O ₃ /SAPO-34	400	1.5	6000	NA	35.0	80.0	3.0	NA	68.0	NA	[12]
In ₂ O ₃ -ZrO ₂ /SAPO-5	300	3	4000	NA	6.7	56.0	3.0	NA	83.0	~14.0	[13]
ZnCr ₂ O ₄ /ZSM-22	360	5	1200	6	21.9	57.6	7.5	48.4	25.6	13.3	[14]

Note: * = CO₂-FTS reaction; ** = catalytic cracking reaction; NA = not available.

Table S17. Elemental composition of the fresh and spent 1.5Na-Fe1Zn1 catalysts.

Catalysts	Fe (wt%)	Zn (wt%)	Na (wt%)
1.5Na-Fe1Zn1-fresh	44.5	25.5	1.7
1.5Na-Fe1Zn1-20h	46.8	27.3	2.1
1.5Na-Fe1Zn1-1000h	42.5	29.0	1.8

Table S18. Iron phase composition of the spent 1.5Na-Fe1Zn1 catalyst after 1000 h CO₂-FTS reaction from Mössbauer spectroscopy.

Catalyst	Phases	Mössbauer parameters				Area (%)
		Hhf (kOe)	IS (mm/s)	QS (mm/s)	Γ/2 (mm/s)	
1.5Na-Fe1Zn1 -1000 h	Fe ₃ O ₄ (A)	487.20	0.27	-0.05	0.16	6.7
	Fe ₃ O ₄ (B)	454.26	0.66	-0.02	0.36	23.3
	χ-Fe ₅ C ₂ (I)	217.05	0.25	-0.10	0.19	27.8
	χ-Fe ₅ C ₂ (II)	183.74	0.18	-0.04	0.19	25.1
	χ-Fe ₅ C ₂ (III)	108.39	0.21	-0.07	0.21	17.1

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